

Sheth L.U.J. And Sir M.V. College

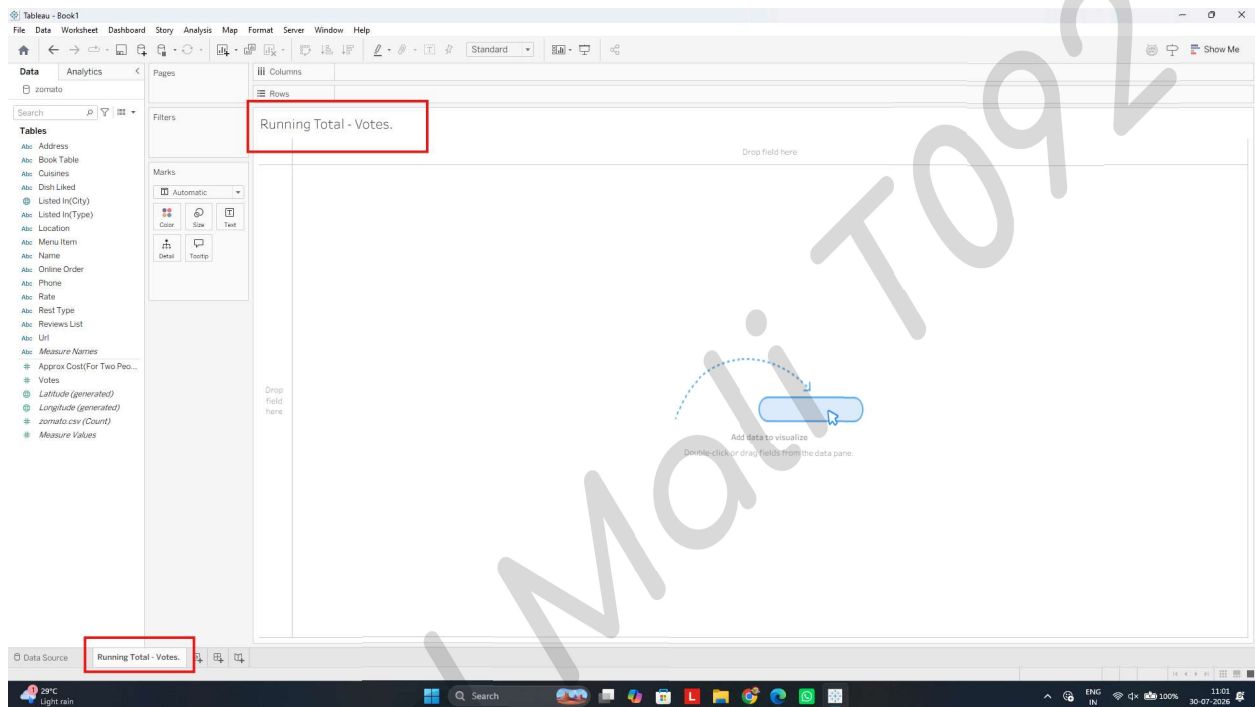
Subject :- Data Visualization with Power BI and Tableau
Practical No 5

Aim:-Analyzing Data using Table Calculations

Task 1: Create a Running Total of Votes by City

A Running Total calculates a cumulative sum across a dimension (e.g., adding each city's votes to the running sum of all previous cities).

1. Open a new worksheet and rename it: Running Total - Votes.

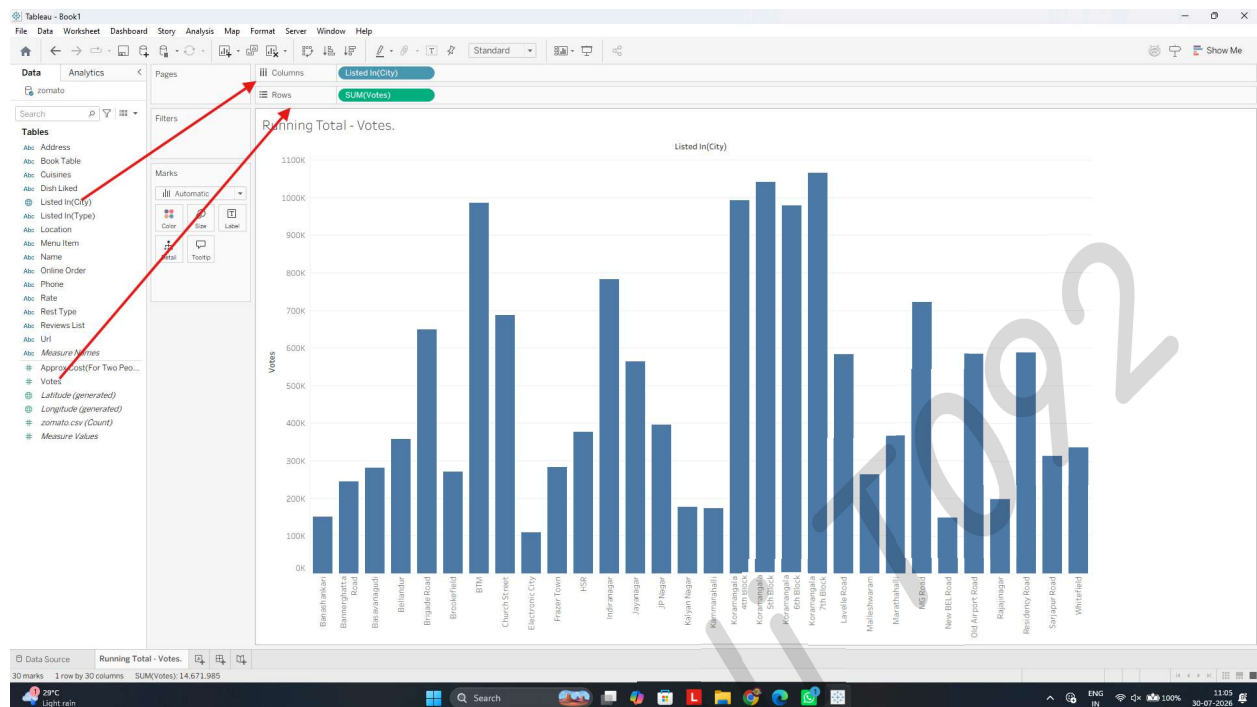


2. From the Data Pane, drag listed_in(city) to the Rows shelf.

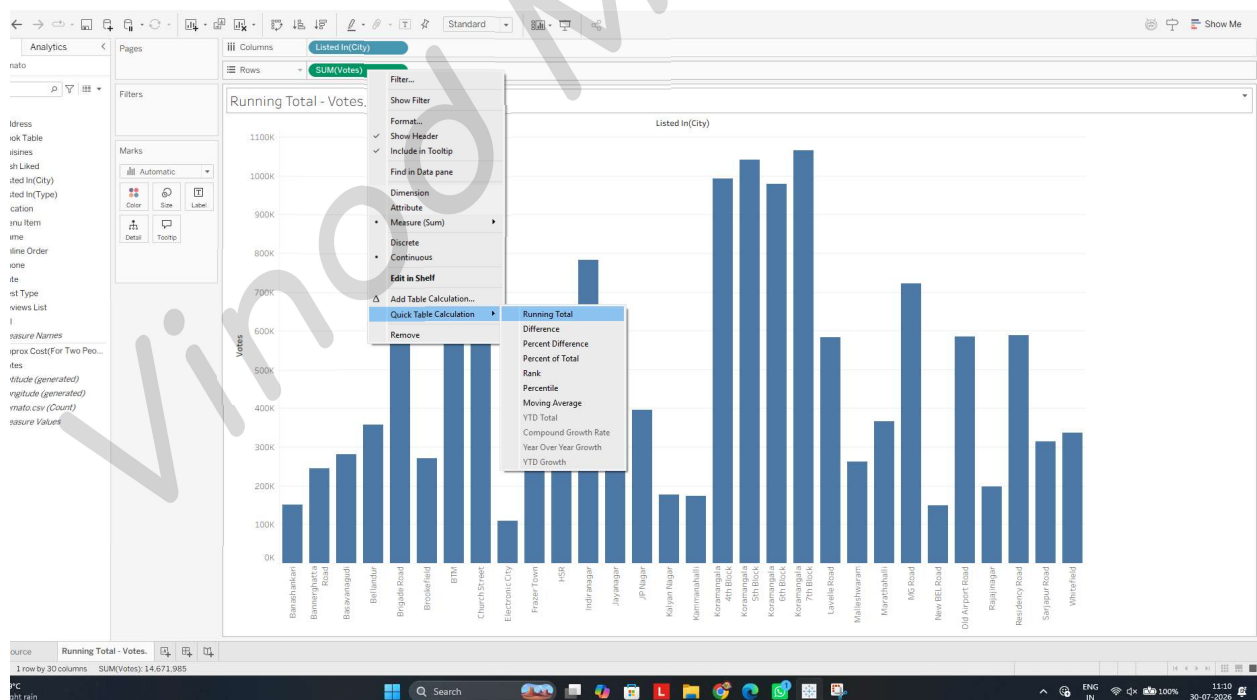
3. Drag votes to the Columns shelf. (Tableau will automatically aggregate it as SUM(votes)).

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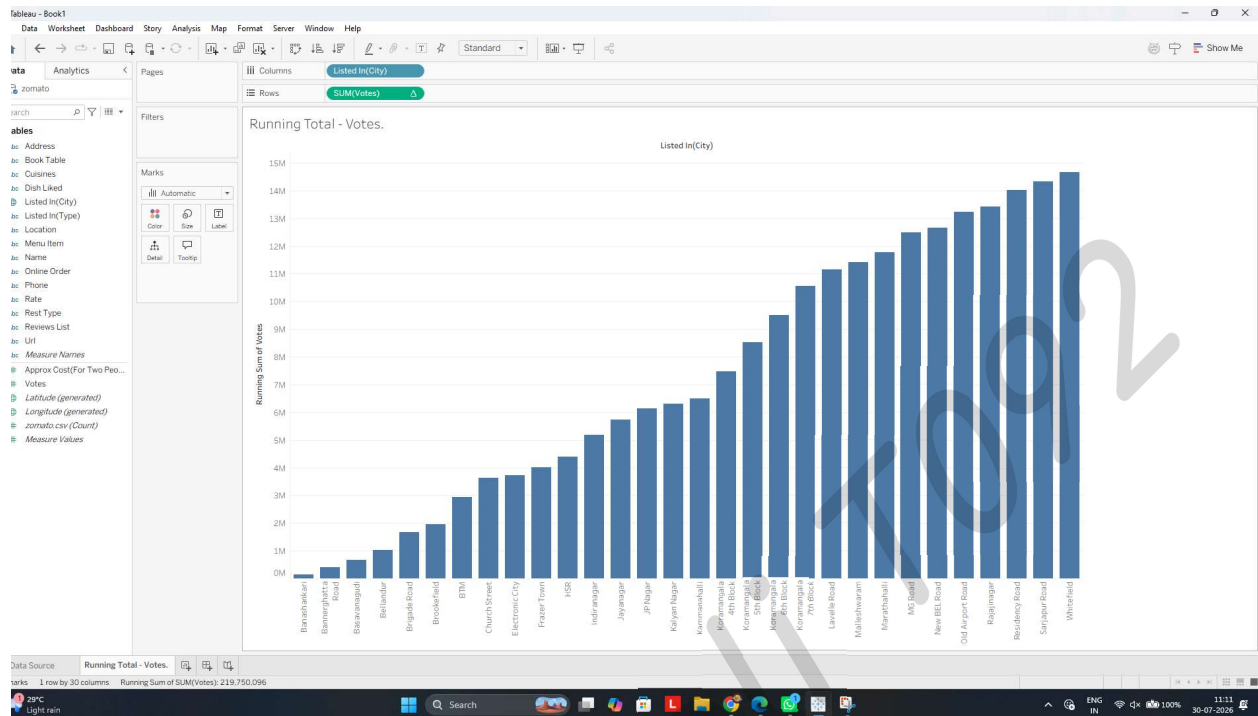


4. Convert to Quick Table Calculation:
- o Right-click the SUM(votes) pill on the Columns shelf.
 - o Hover over Quick Table Calculation and select Running Total.



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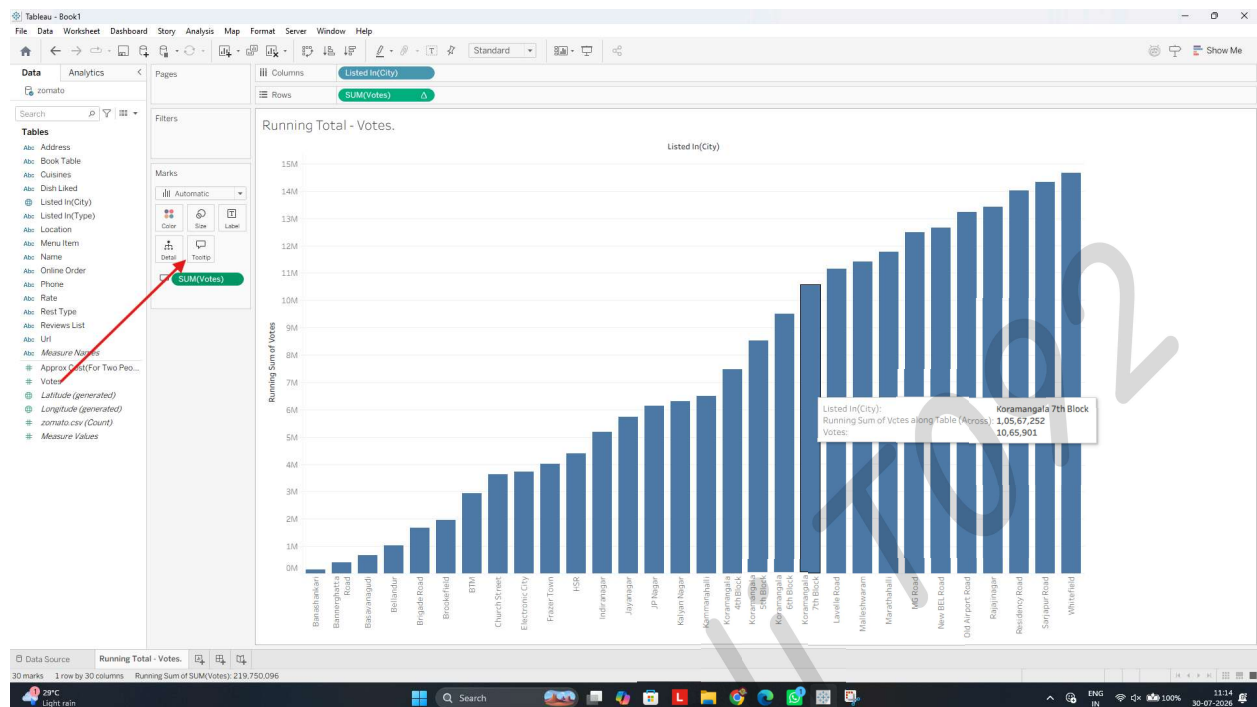


5. Add Visual Context (Secondary Measure):

- o Drag votes from the Data Pane again and drop it directly onto the Tooltip mark in the Marks Card.
- o Result: Now when you hover over a bar, you will see both the individual city's vote count and the cumulative total up to that city.

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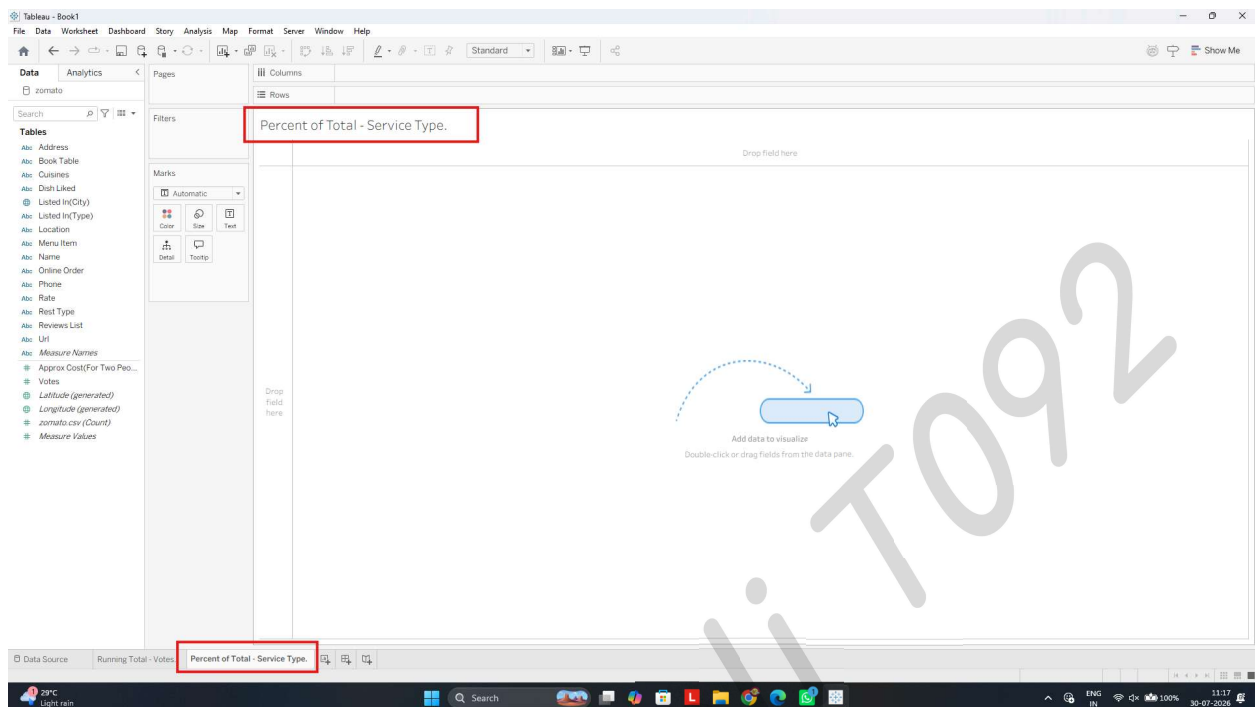
Task 2: Create a Percent of Total across Categories

A Percent of Total transforms row numbers into relative percentages of the entire view (summing to 100%).

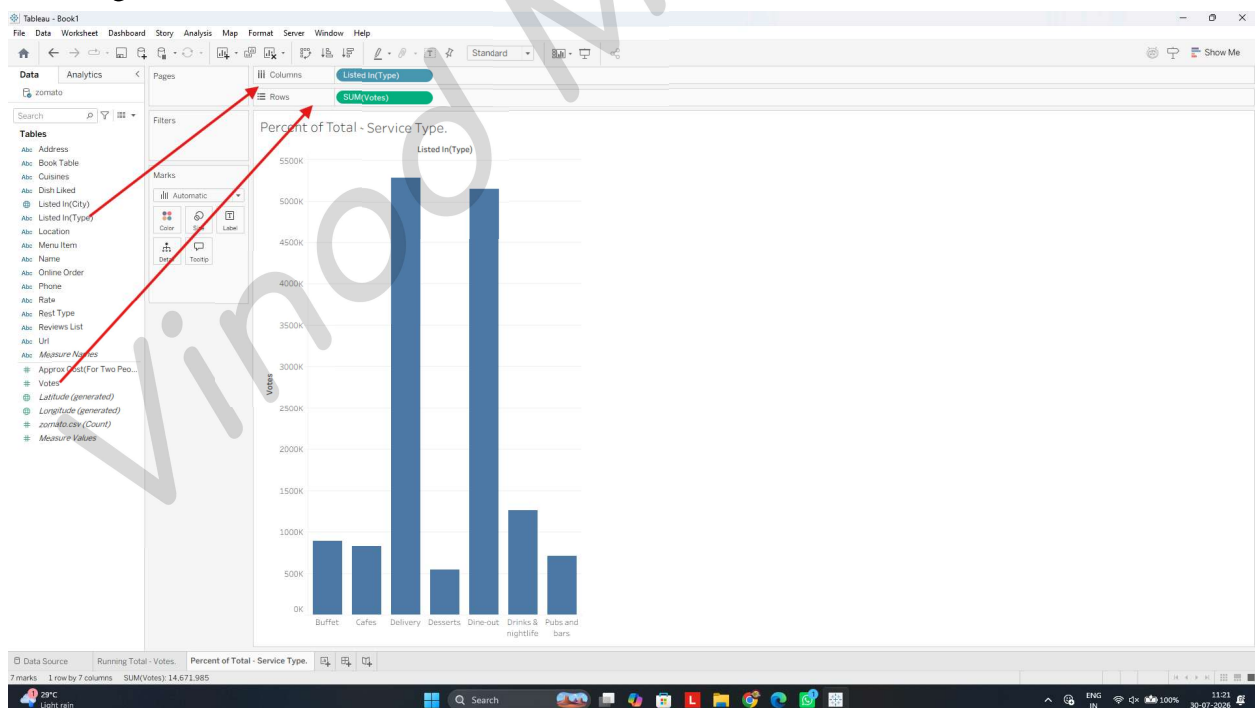
1. Open a new worksheet and rename it: Percent of Total - Service Type.

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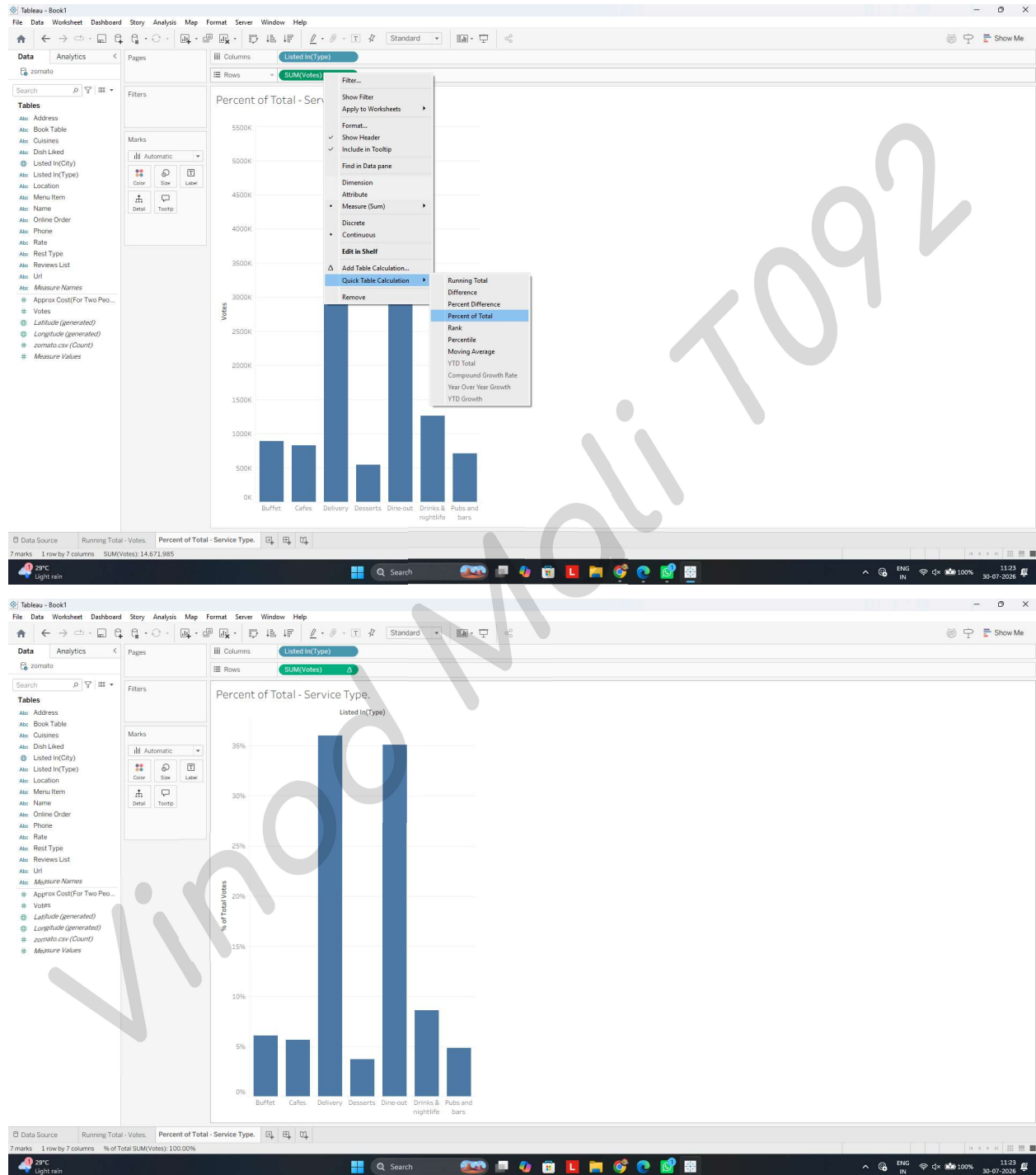
2. Drag listed_in(type) (e.g., Buffet, Cafes, Delivery, Dining) to the Rows shelf.
3. Drag votes to the Columns shelf.



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4. Right-click the SUM(votes) pill on the Columns shelf.
5. Hover over Quick Table Calculation and select Percent of Total.

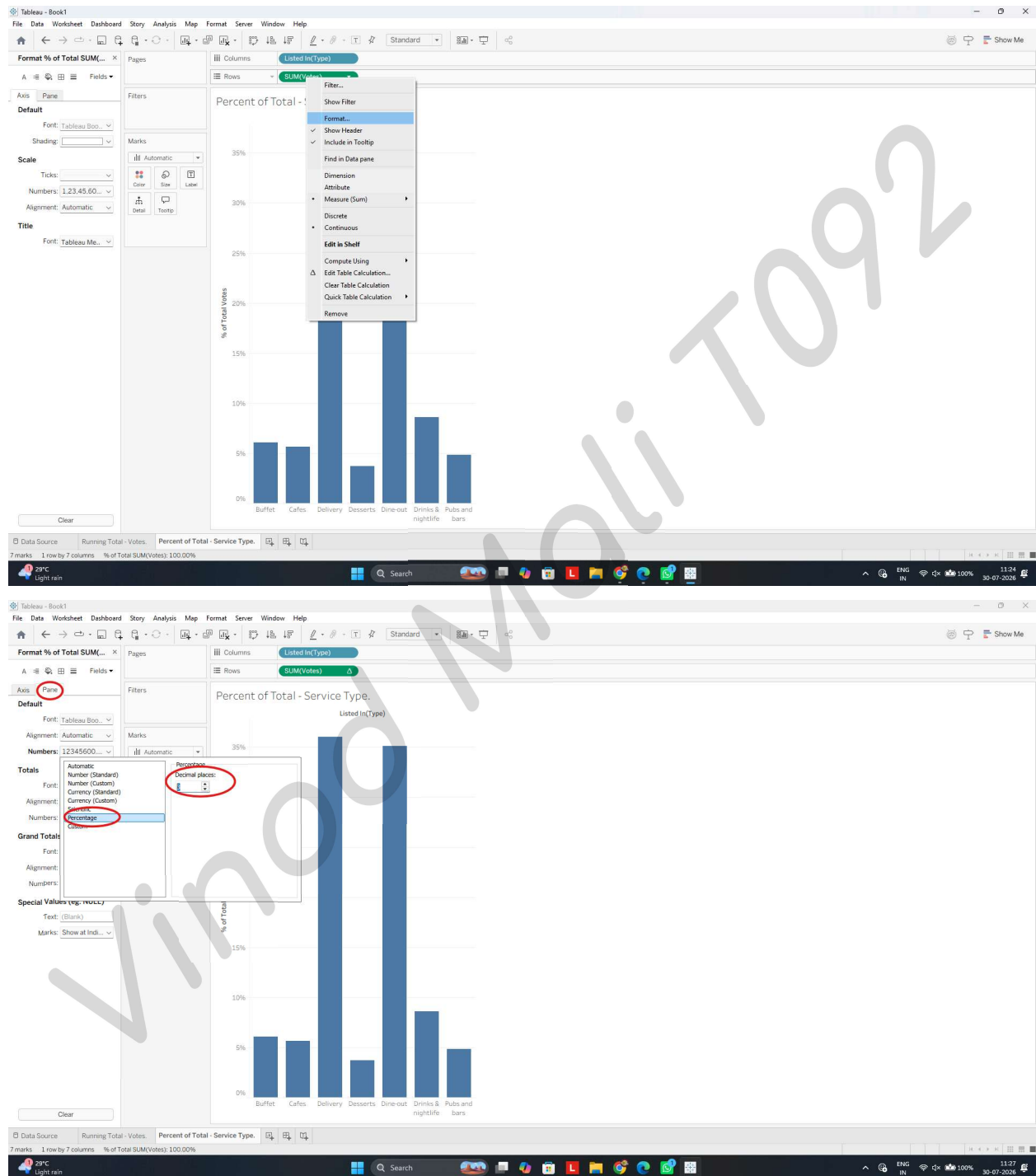


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6. Format as Percentage:

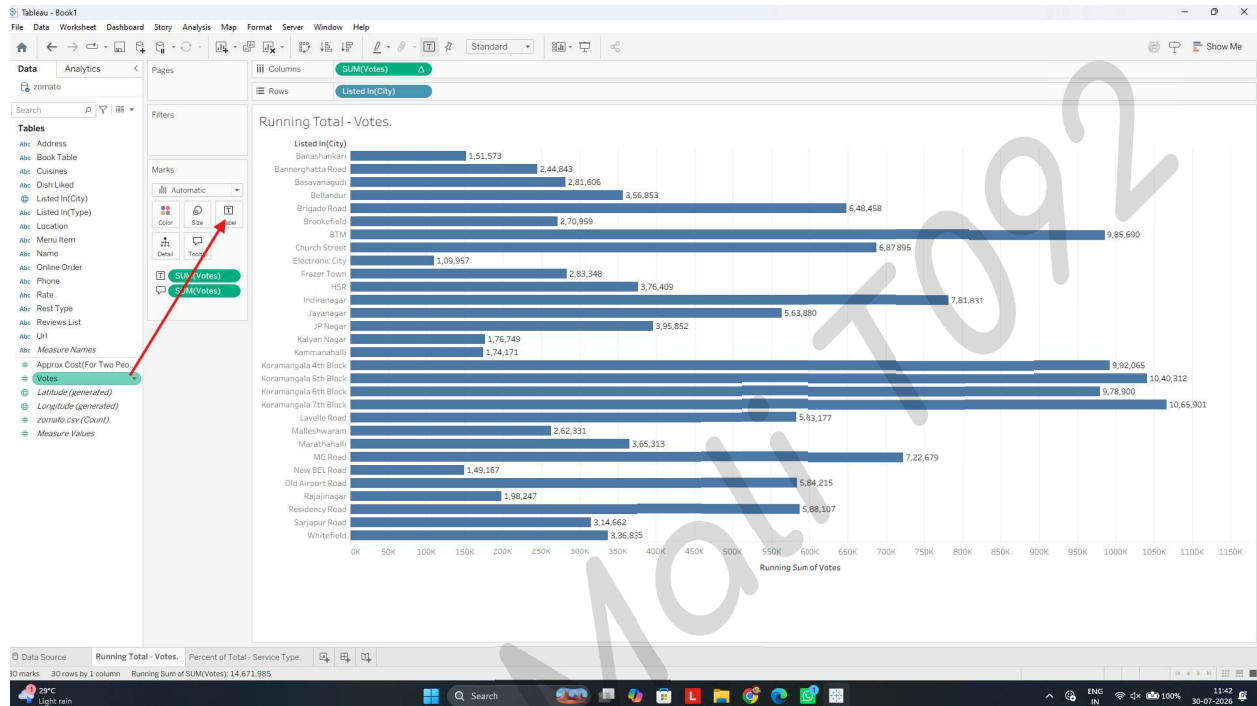
o Right-click the SUM(votes) pill again and select Format...



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7. Drag SUM(votes) (with the table calculation icon Δ) onto the Label mark so the exact percentages display on top of each bar. (NOTE: IF YOU WANT TO SEE THE BAR GRAPH TOO, YOU CAN DRAG SUM(votes) TO LABEL ICON BY HOLDING CTRL KEY)

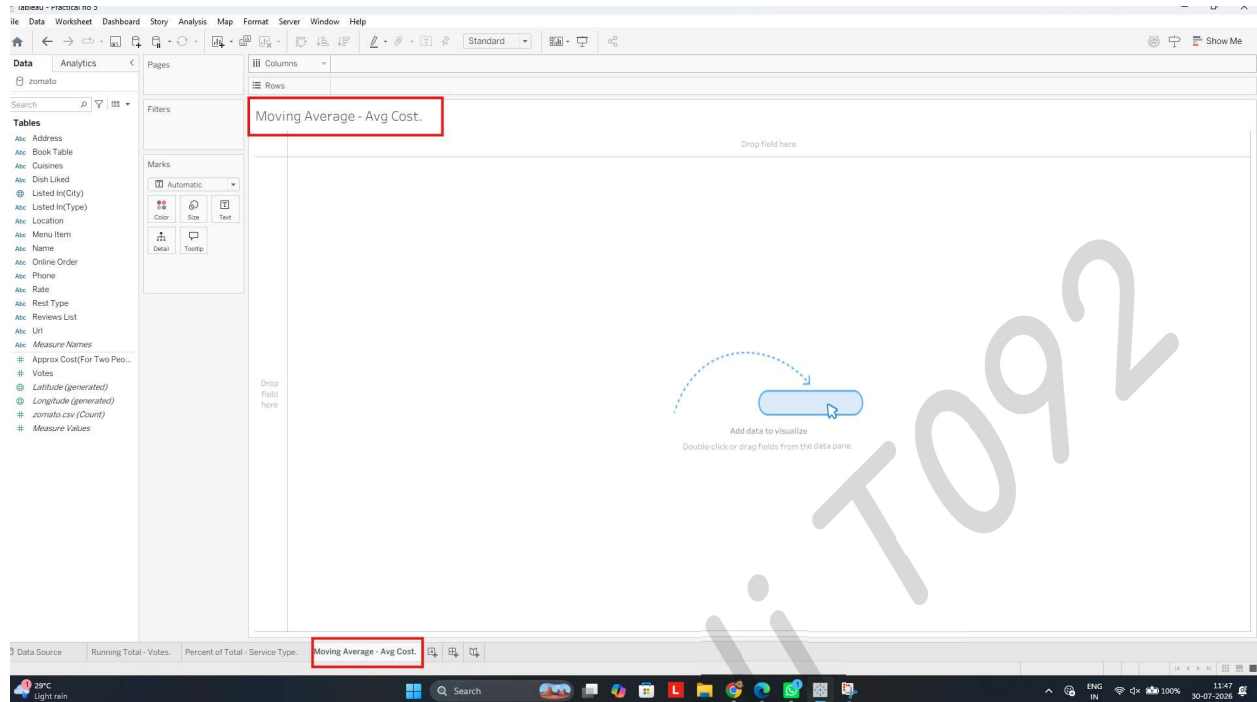


Task 1: Create a Moving Average Calculation

We will create a moving average to smooth out variations in pricing across restaurants based on vote counts or listing order.

1. Open a new worksheet and rename it: Moving Average - Avg Cost.

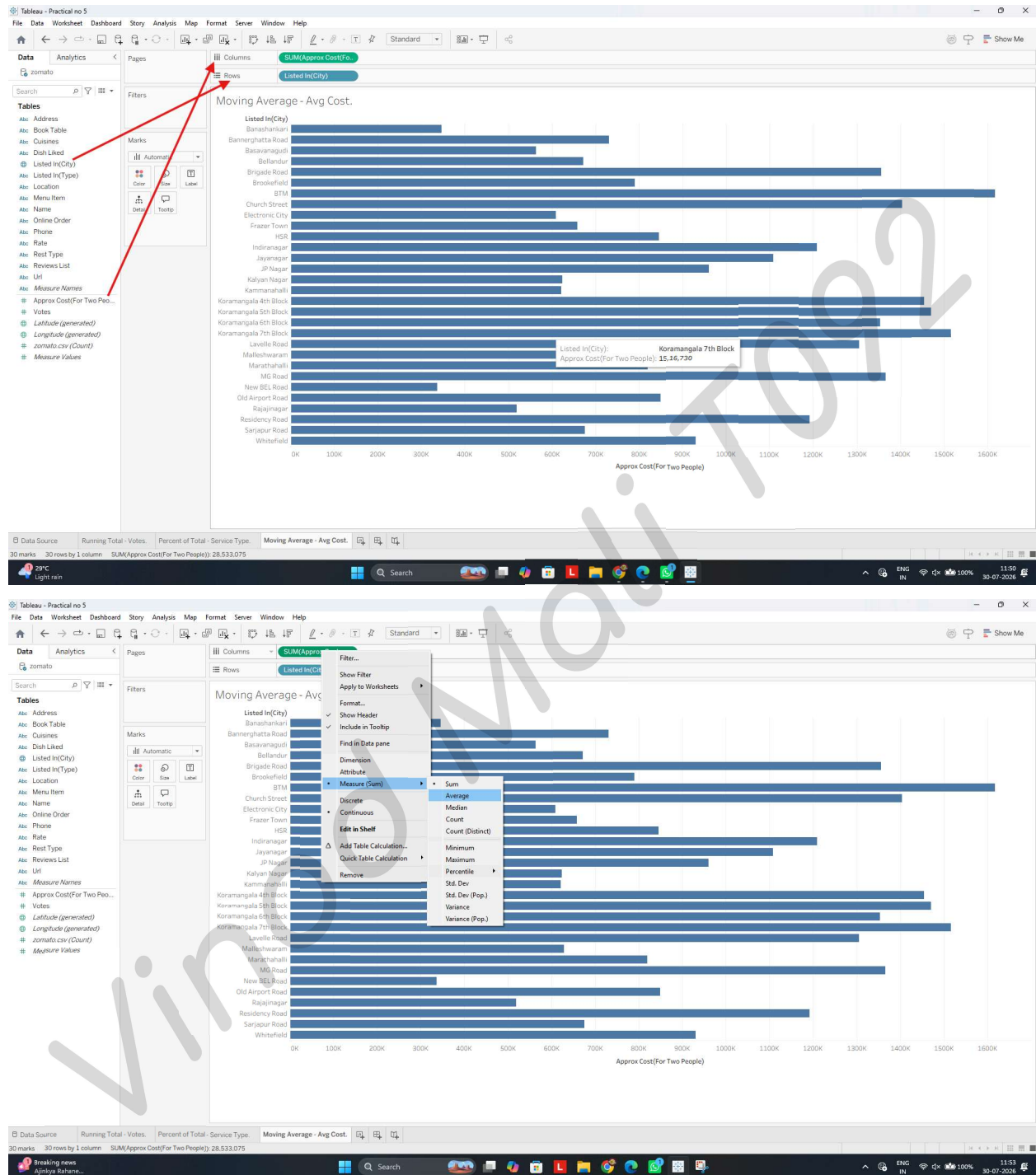
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2. Drag listed_in(city) to the Rows shelf.
3. Drag approx_cost(for two people) to the Columns shelf. (Note: If Tableau imports this field as a String/Dimension due to commas, right-click the field in the Data Pane, select Change Data Type Number (whole), then drag it as AVG or after placing it on column it doesn't shows AVG right click on it Hover at Measure and click Average).

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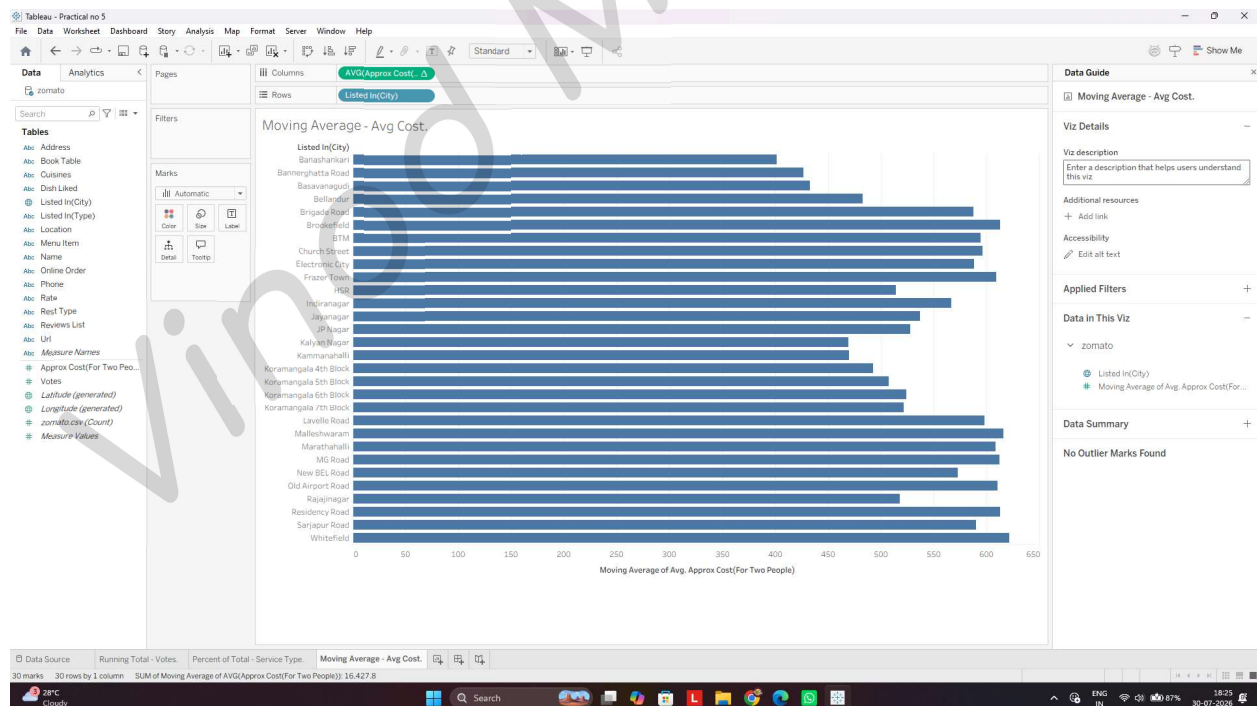
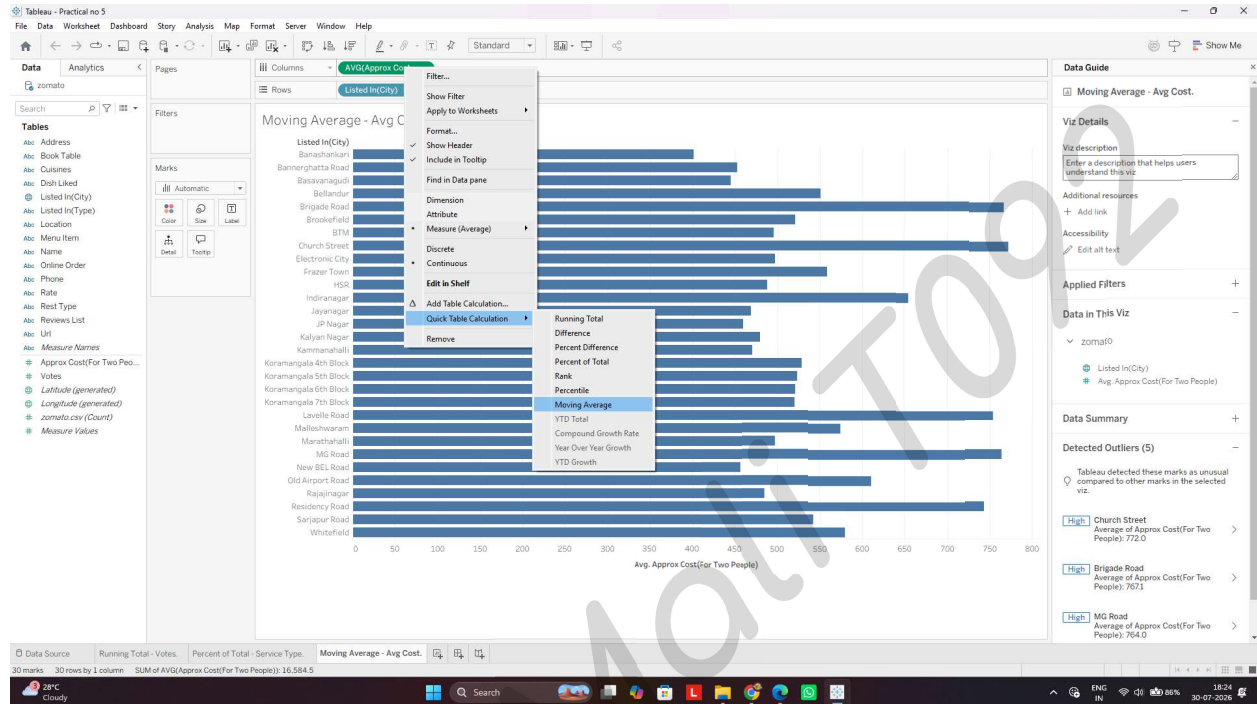


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4. Convert to a Moving Average Quick Table Calculation:

- o Right-click the AVG(approx_cost...) pill on the Columns shelf.
- o Hover over Quick Table Calculation and select Moving Average.

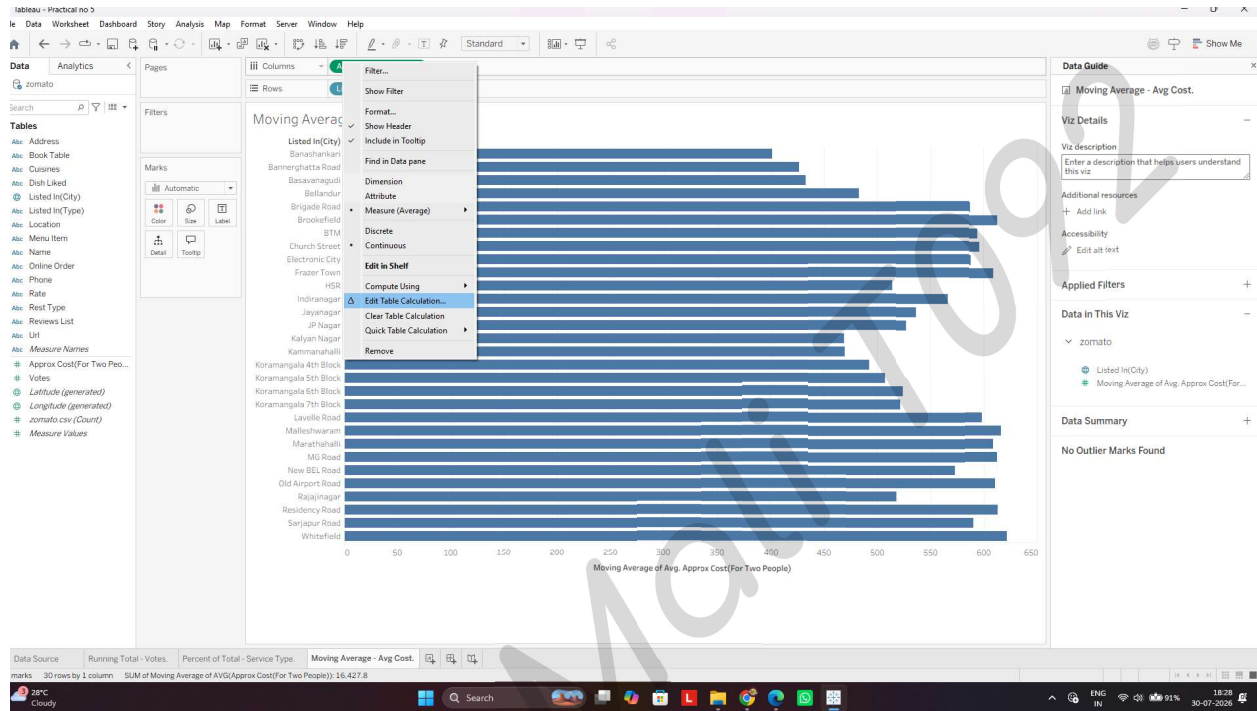


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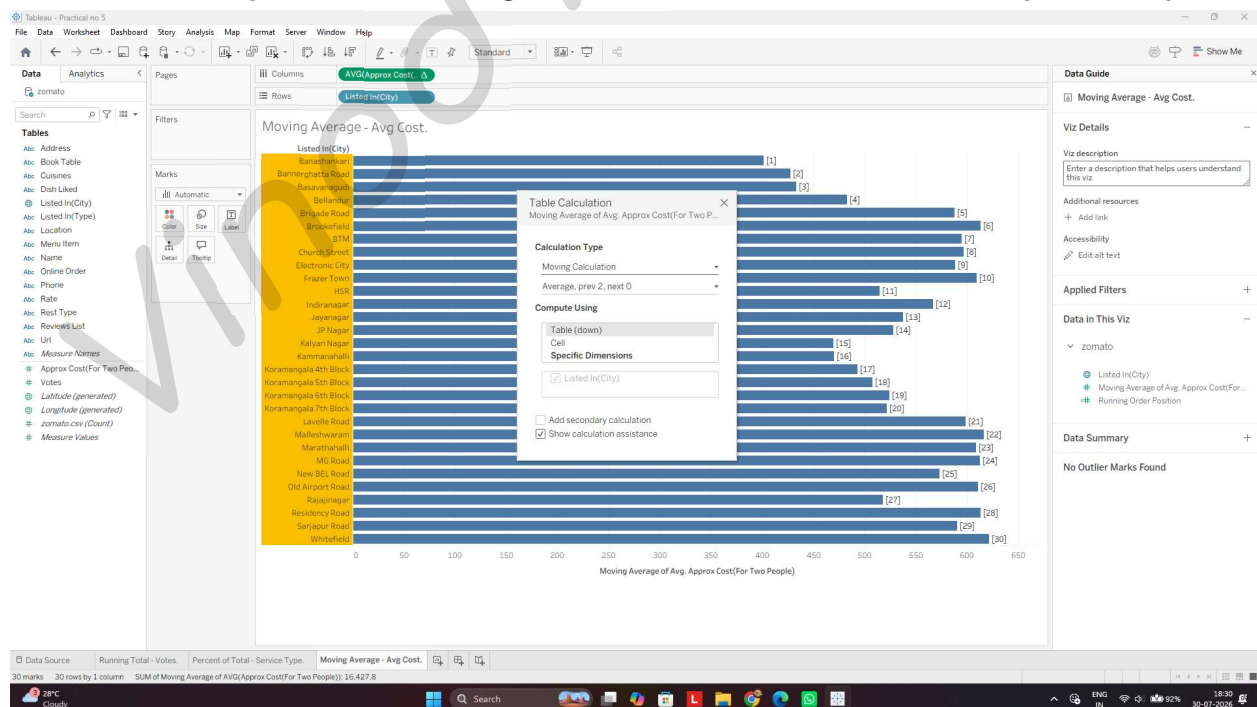
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5. Customize the Moving Window:

- o Right-click the modified AVG(approx_cost...) pill again (it now has a Δ symbol).
- o Select Edit Table Calculation...



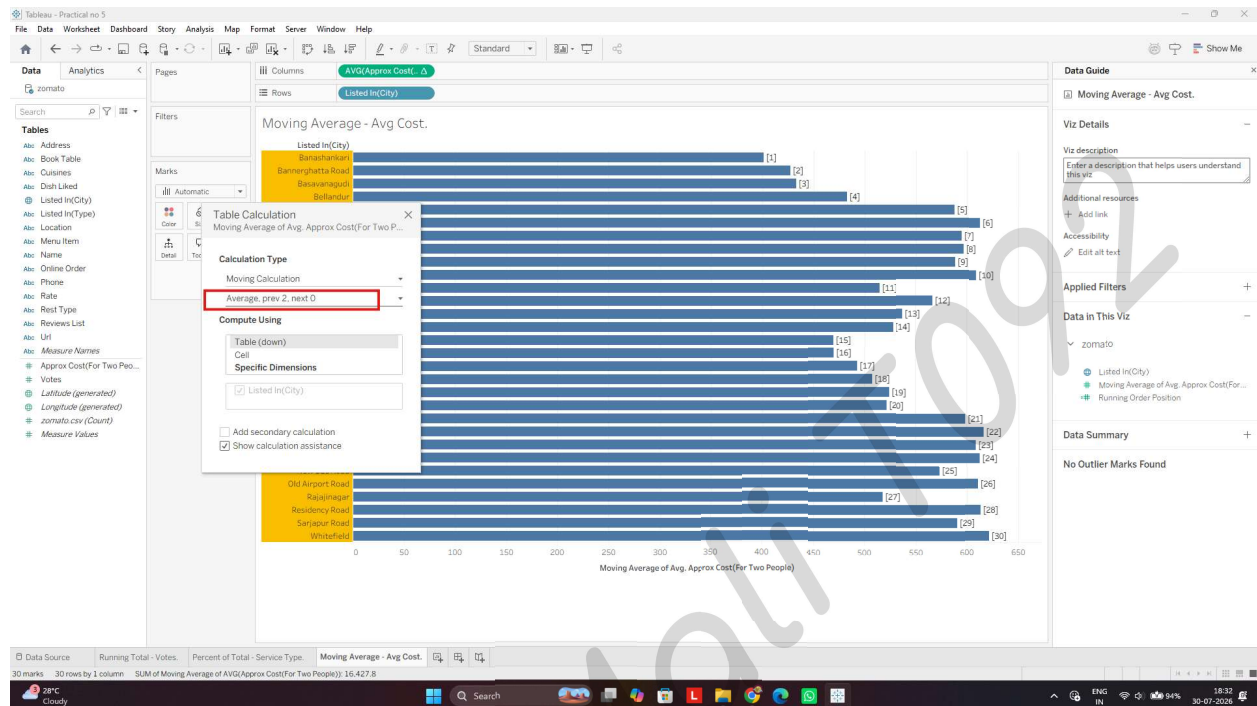
- o In the configuration dialog, set Summarize values using: Average.



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o Set Previous values: 2 and Next values: 0 (This calculates a 3-point moving average including the current city and 2 previous cities).



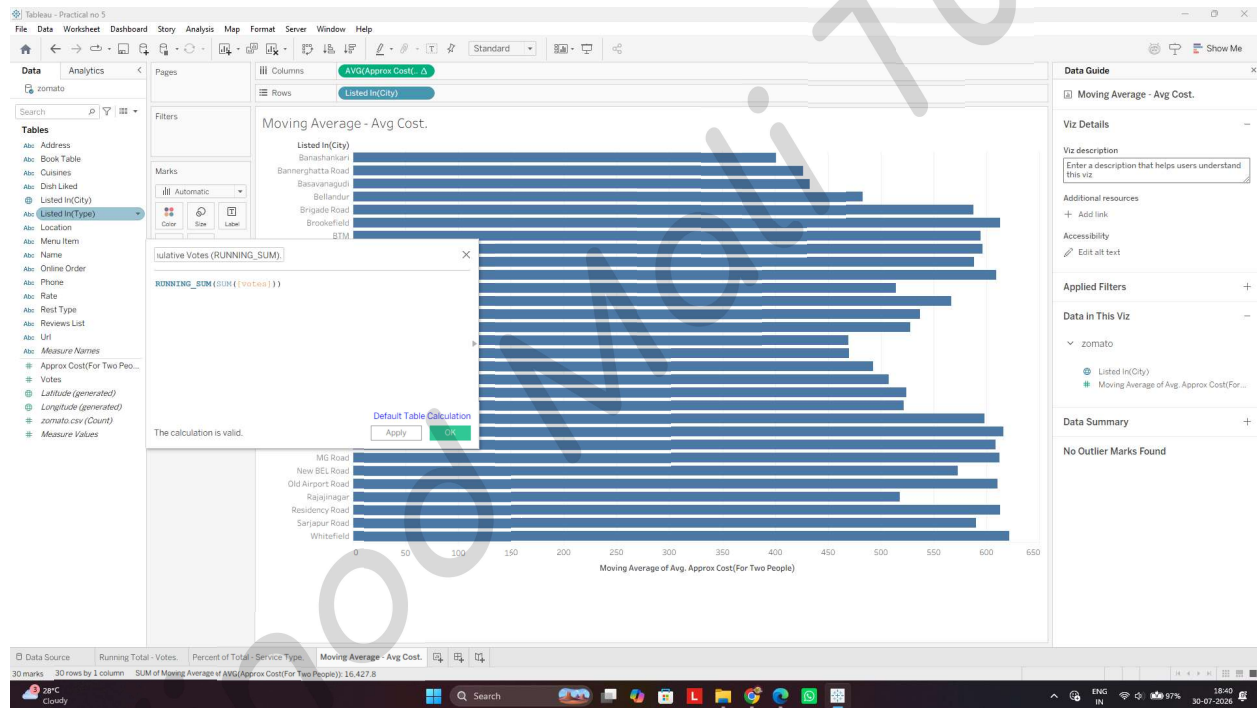
o Close the dialog box.

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Task 2: Create a Custom Cumulative Metric via Calculated Field
Instead of using Quick Table Calculations, we can write a dedicated Table Calculation function formula directly.

1. Click the small drop-down arrow at the top right of the Data Pane (just next to the Search box) and select Create Calculated Field...
Alternative shortcut: Right-click anywhere on a specific field or dimension in the Data Pane Create Calculated Field... (or press Alt + Analysis menu Create Calculated Field).
2. Name the field: Cumulative Votes (RUNNING_SUM).
3. Enter the following Table Calculation formula:
`RUNNING_SUM(SUM([votes]))`



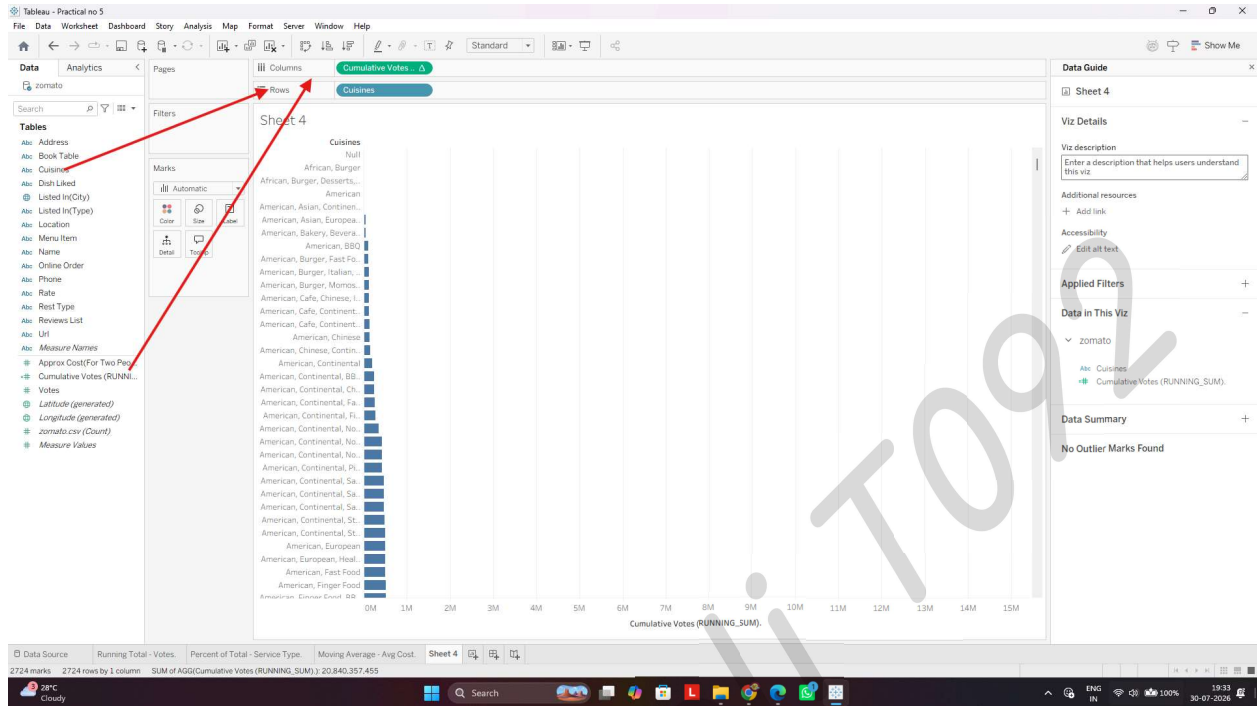
4. Click OK.
5. Open a new worksheet.
6. Drag cuisines to Rows (Filter to top 15 or 20 if the list is too long).
7. Drag your newly created Cumulative Votes (RUNNING_SUM) field onto Columns.

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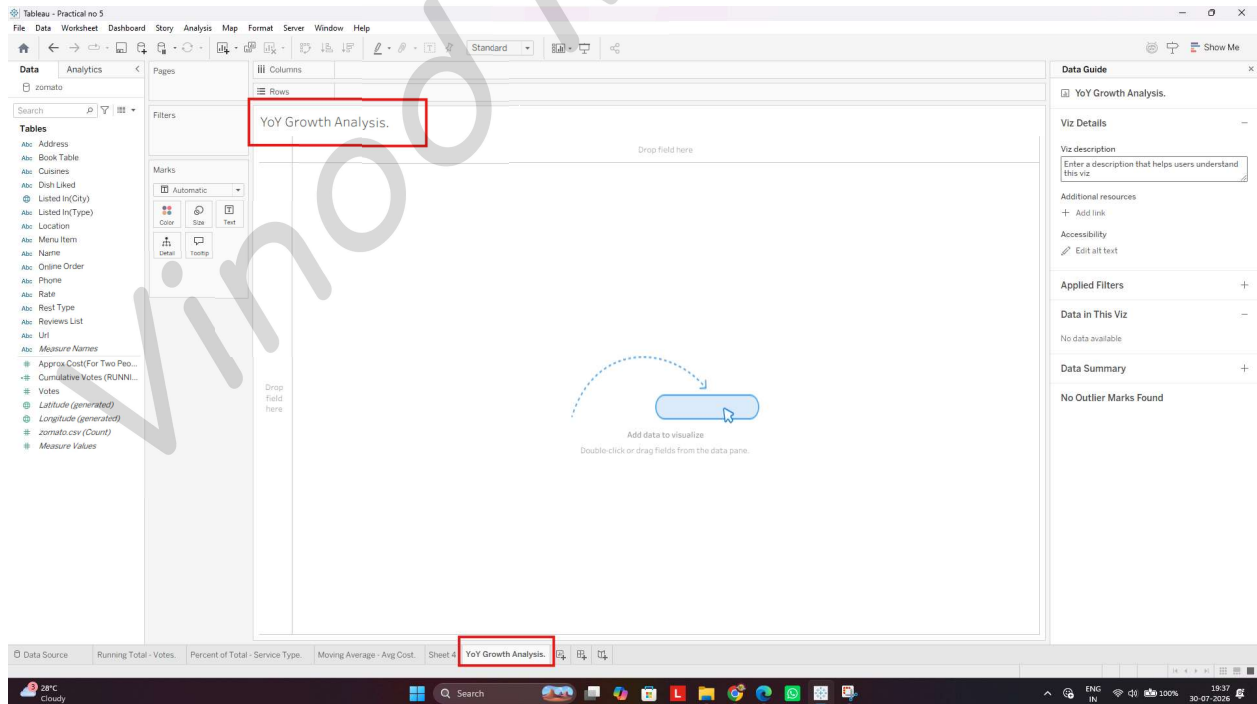
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C: Perform year-over-year growth analysis

Task: Calculate Year-over-Year (YoY) / Period-over-Period Percent Growth

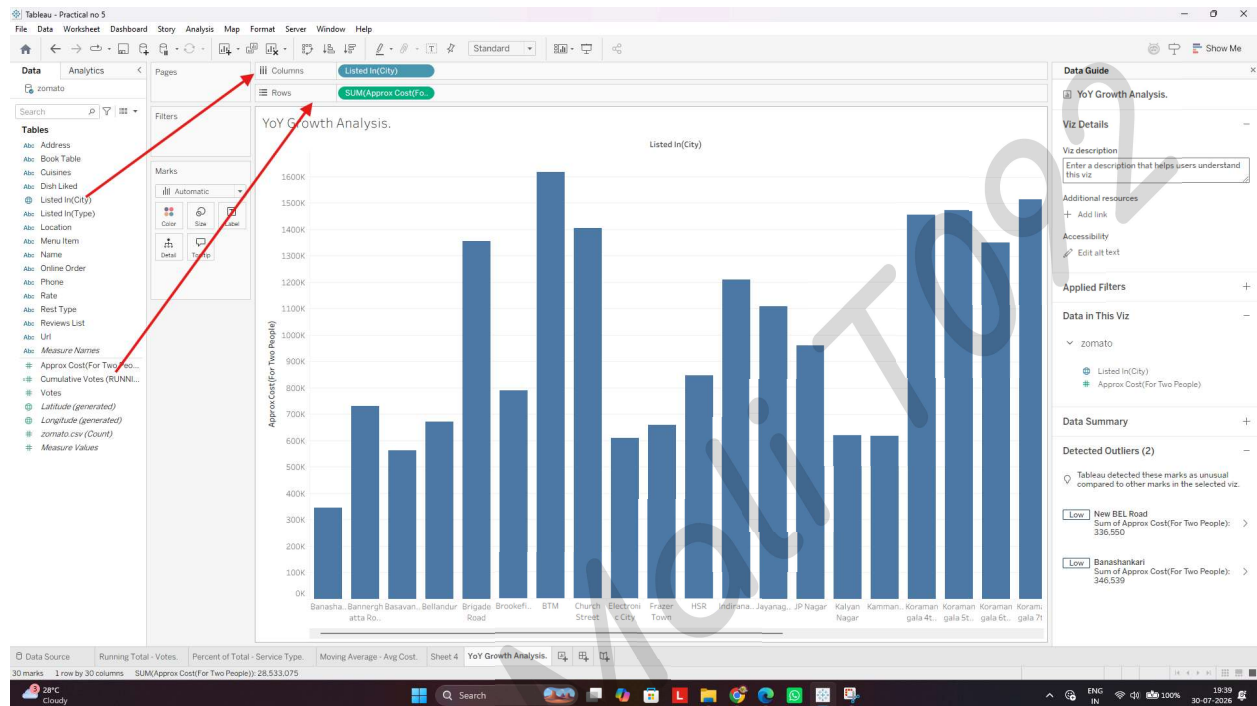
1. Open a new worksheet and rename it: YoY Growth Analysis.



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2. Place your time dimension (or ordinal grouped column like listed_in(city)) on the Columns shelf.
3. Drag votes (or approx_cost(for two people)) onto the Rows shelf as SUM(votes).

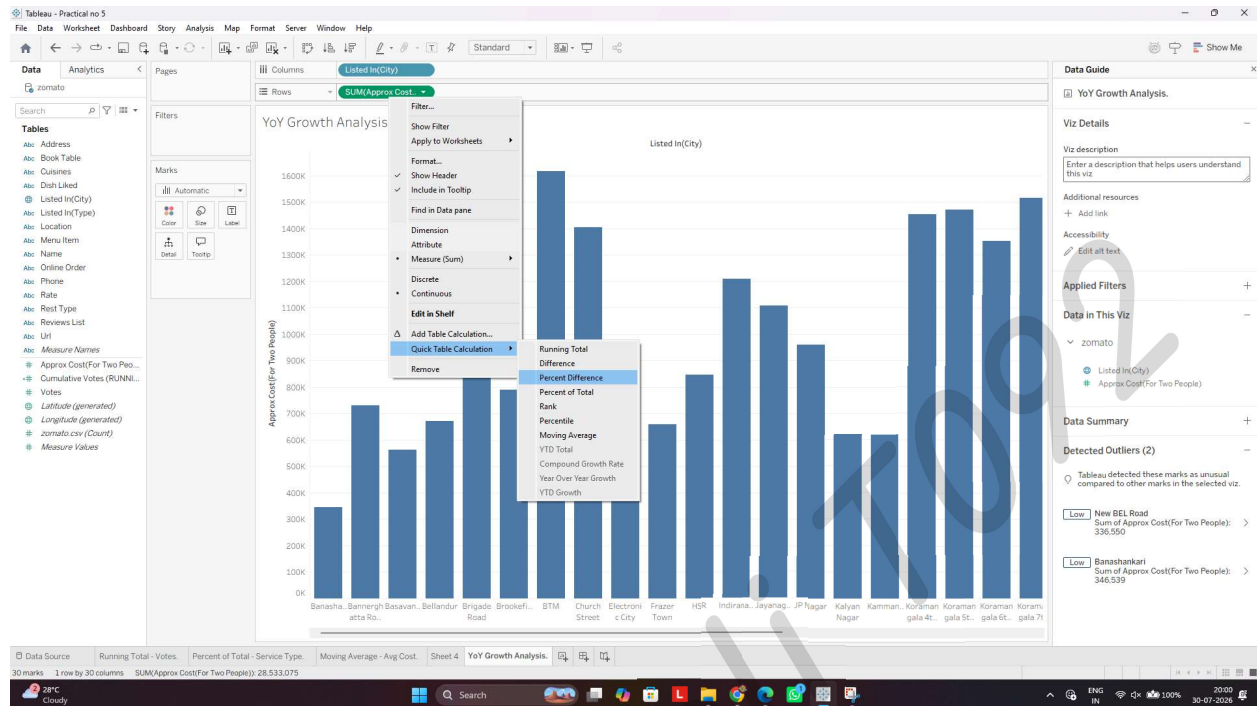


4. Apply the Year-over-Year Growth Quick Table Calculation:
 - o Right-click the SUM(votes) pill on the Rows shelf.
 - o Hover over Quick Table Calculation select Percent Difference (or Year Over Year Growth if using a true Date field).

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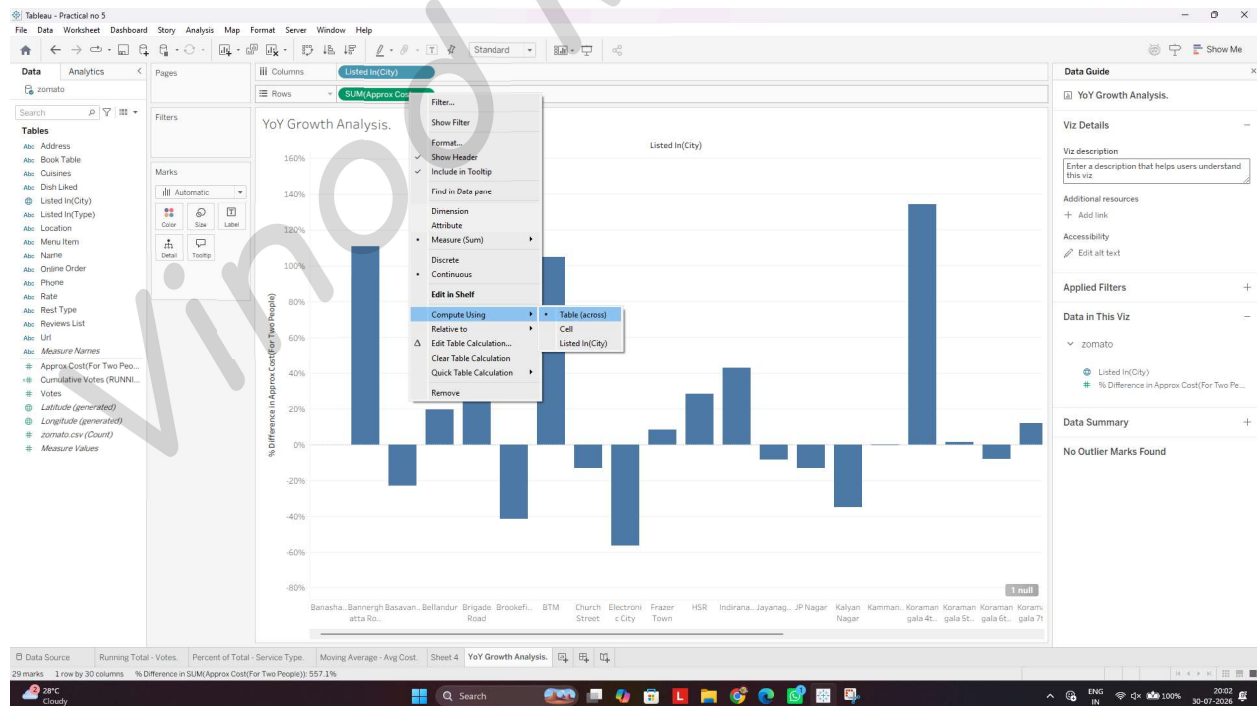
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5. Configure the Growth Calculation Direction:

- o Right-click the modified SUM(votes) pill (with the Δ icon).
- o Hover over Compute Using select Table (across){In some cases it may have by selected by default}.



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FileDataWorksheetDashboardStoryAnalysisMapFormatServerWindowHelp

DataAnalytics

zomato

Search

Tables

Address

Book Table

Cuisines

Dish Liked

Listed In(City)

Listed In(Type)

Location

Menu Item

Name

Online Order

Phone

Rate

Rest Type

Reviews List

URI

Measure Names

Approx Cost(For Two Perso...

Cumulative Votes (RUNN...

Votes

Latitude (generated)

Longitude (generated)

zomato.cov (Client)

Measure Values

Columns

Listed In(City)

Rows

SUM(Approx Cost...)

Filters

None

Visuals

Automatic

Color

Size

Label

Detail

Tooltip

YoY Growth Analysis.

Listed In(City)

% Difference in Approx Cost(For Two People)

Bar chart showing YoY Growth Analysis for various locations. The Y-axis represents the percentage difference in approximate cost for two people, ranging from -80% to 160%. The X-axis lists various locations. The bars show varying growth rates, with a significant peak for 'Koraman Korman' and a significant dip for 'Koraman Korman gala 4t...'.

Data Source

Running Total - Votes

Percent of Total - Service Type

Moving Average - Avg Cost

Sheet 4

YoY Growth Analysis

Data Guide

YoY Growth Analysis.

Viz Details

Viz description

Additional resources

Accessibility

Applied Filters

Data in This Viz

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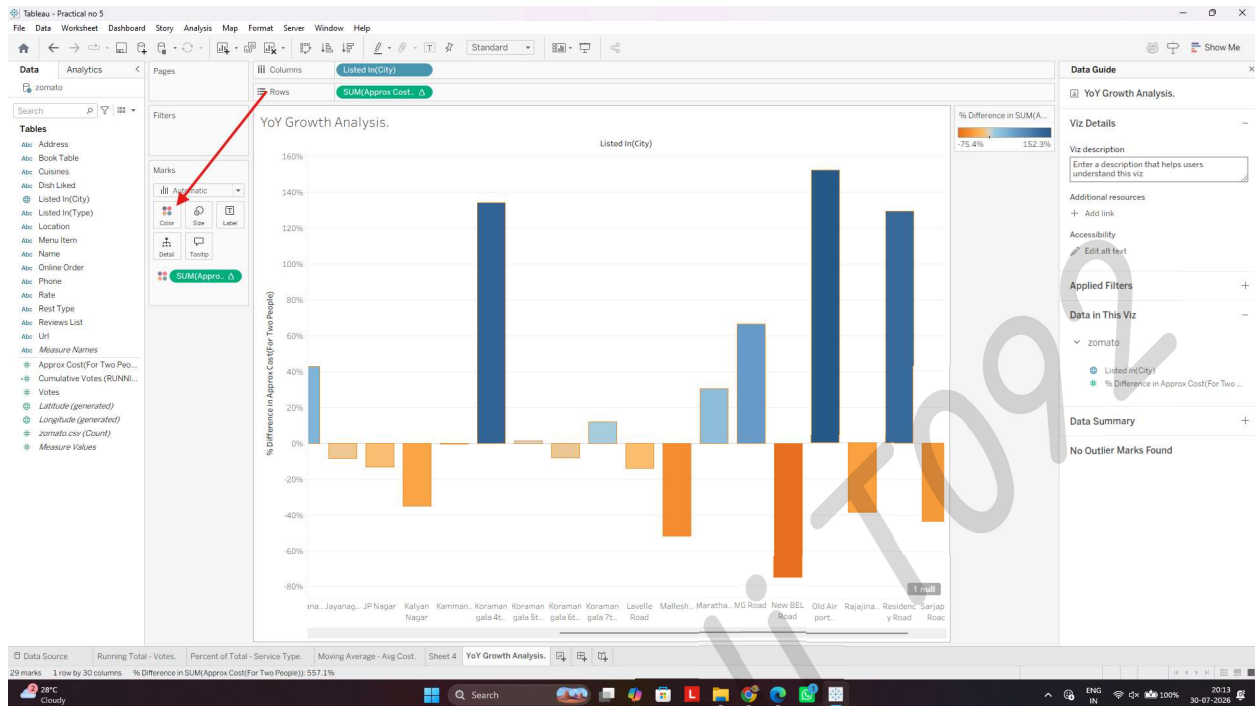
Data Summary

No Outlier Marks Found

- o Hold down Ctrl (or Cmd on Mac), click and drag the SUM(votes) pill from the Rows shelf, and drop it directly onto the Color mark on the Marks card.
- o Positive growth will appear in one color (e.g., Green/Blue) while negative performance dips will turn Red/Orange!

Add Visual Color Indicators:
Hold down Ctrl (or Cmd on Mac), click and drag the SUM(votes) p from the Rows shelf, and drop it directly onto the Color mark on the Marks card.
Positive growth will appear in one color (e.g., Green/Blue) while negative performance dips will turn Red/Orange!

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D: Use ranking functions (Top N, Bottom N)

Ranking table calculations evaluate each row's aggregate value relative to all other rows in the partition, assigning a numeric rank (1, 2, 3...).

Task 1: Create a Rank Calculation

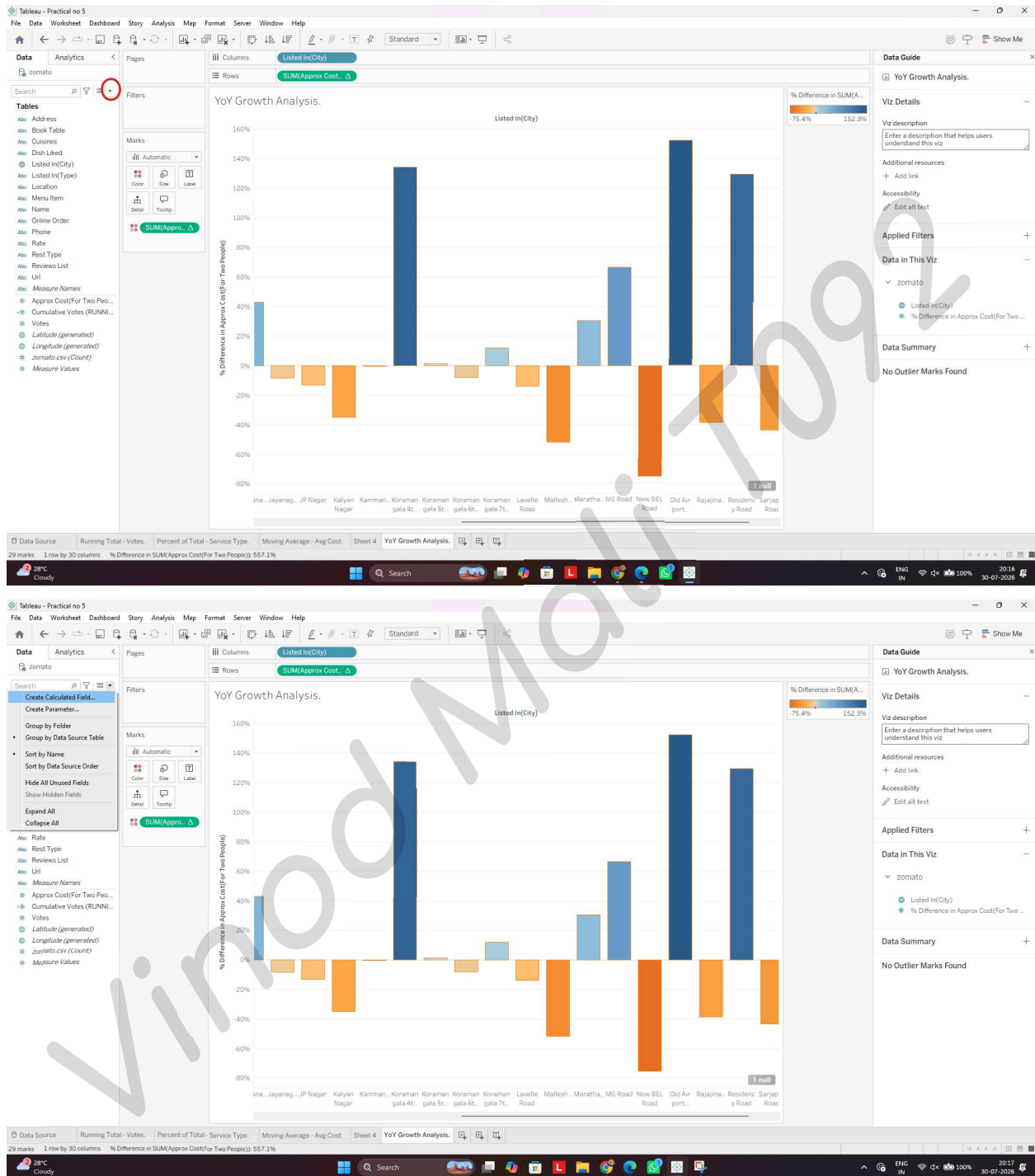
1. Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field...

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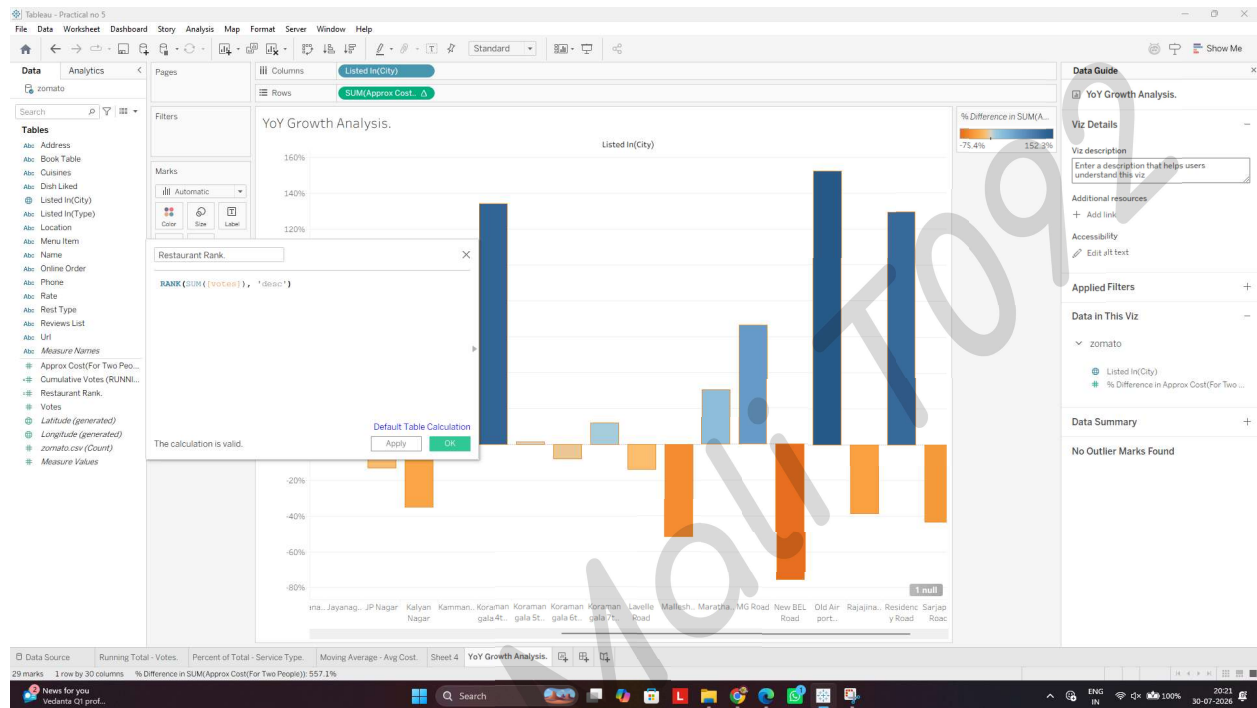


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2. Name the field: Restaurant Rank.
3. Input the following ranking calculation:

`RANK(SUM([votes]), 'desc')`



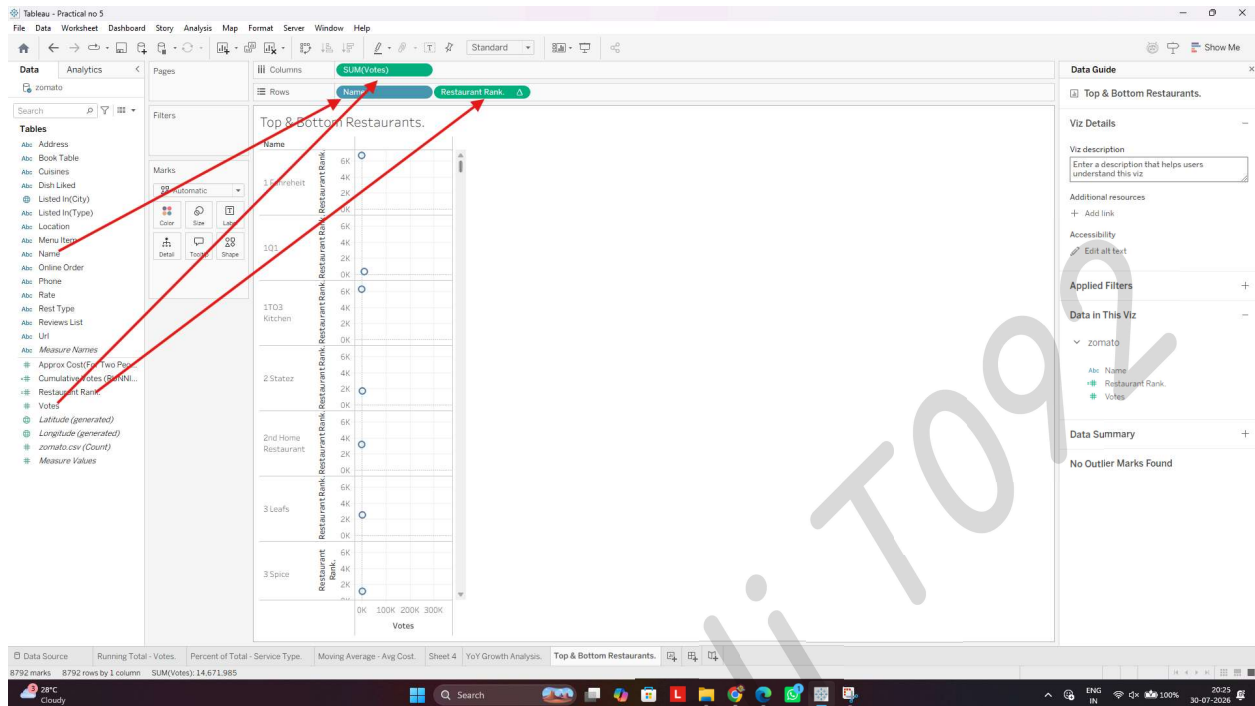
4. Click OK. Task

2: Build the Top N / Bottom N Ranked View

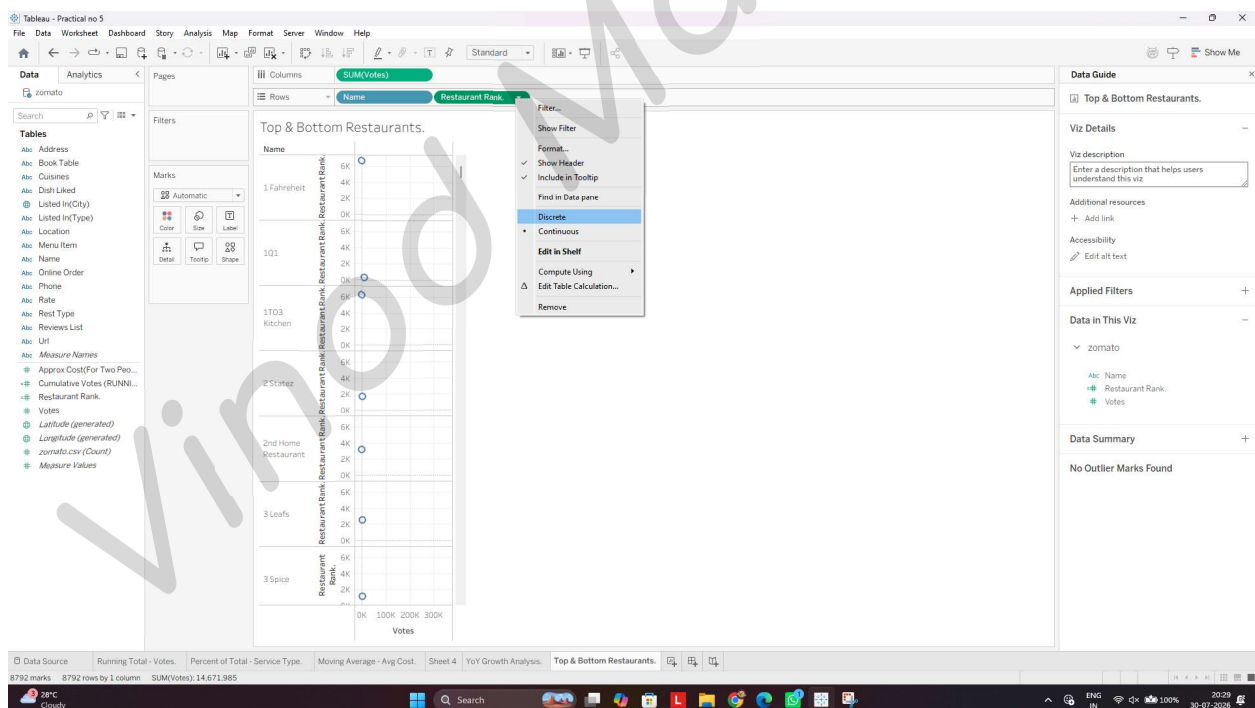
1. Open a new worksheet and rename it: Top & Bottom Restaurants.
2. Drag name (Restaurant Name) to the Rows shelf.
3. Drag votes to the Columns shelf.
4. Drag your newly created Restaurant Rank calculated field onto the Rows shelf, placing it to the left of name.

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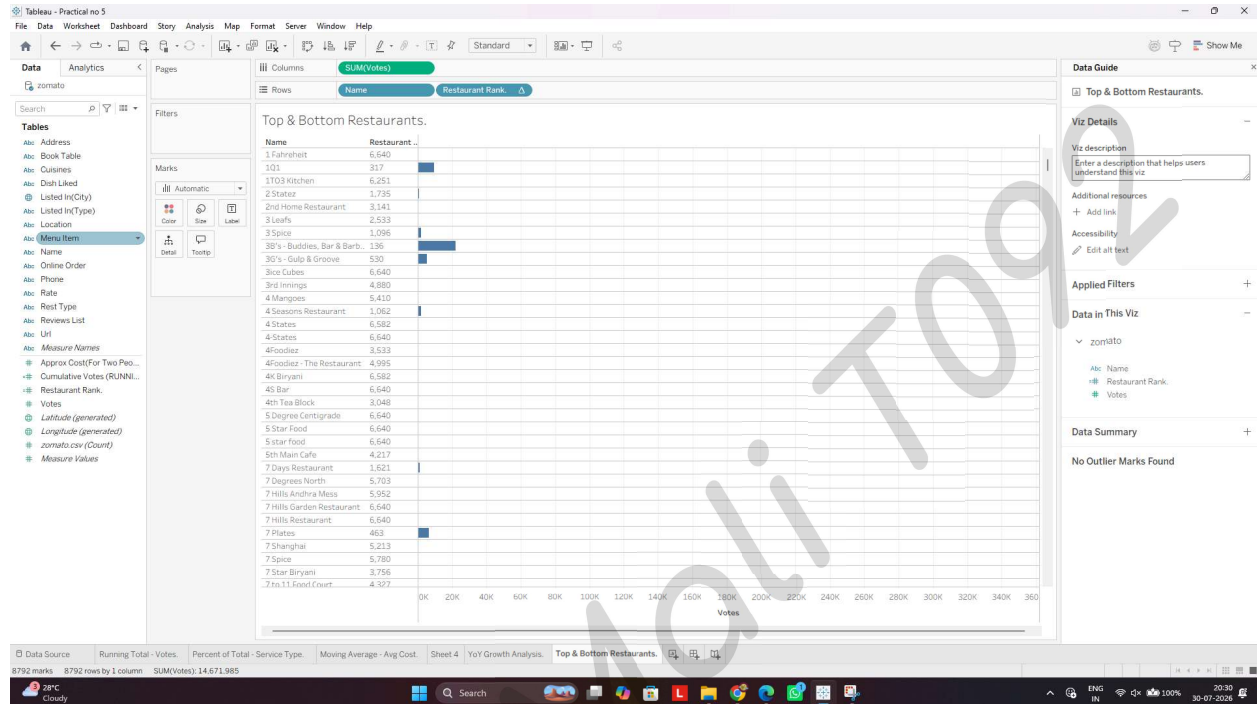
5. Convert Restaurant Rank to a Discrete field:



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o Right-click the Restaurant Rank pill on the Rows shelf and select Discrete. (The pill will turn blue, ordering the restaurants automatically from #1 onward).

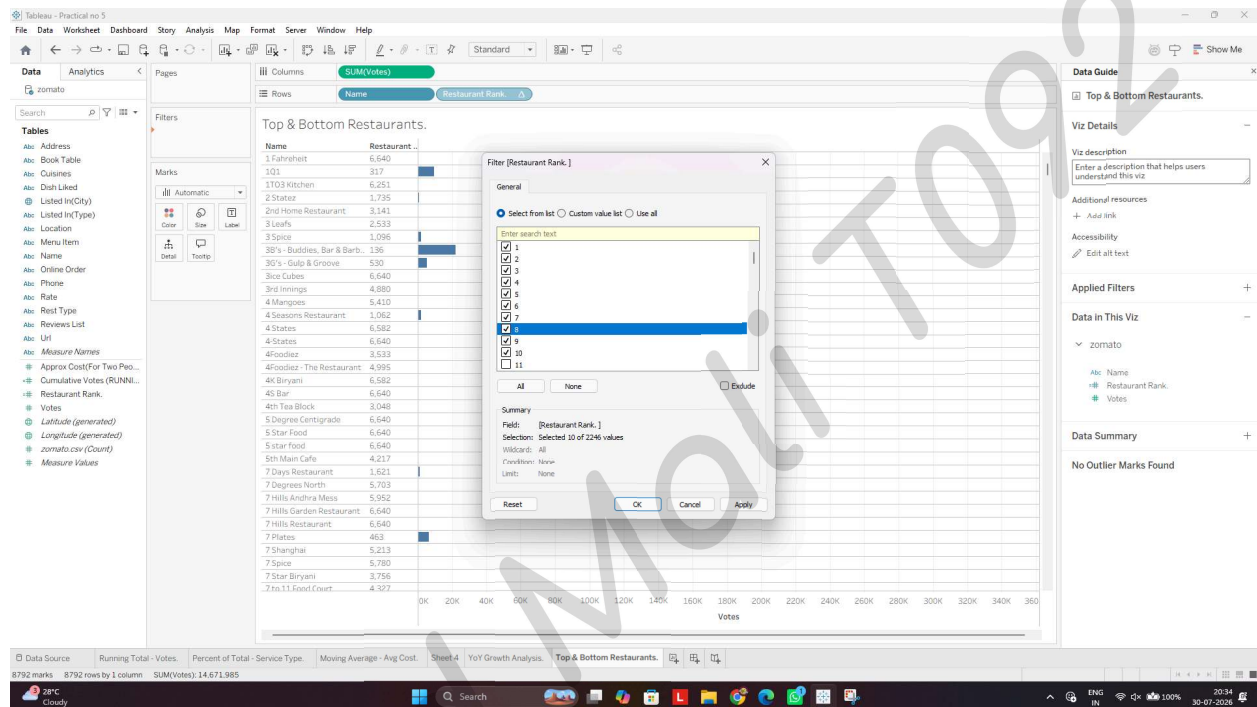


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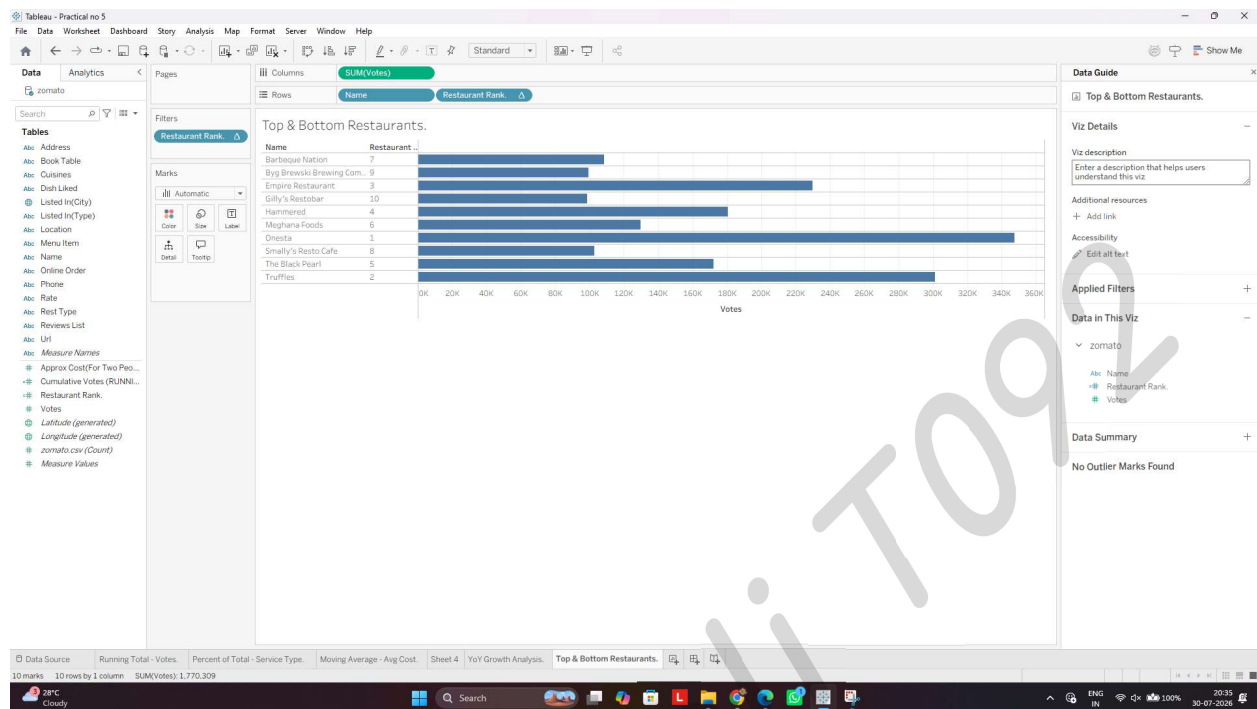
Task 3: Filter Dynamically to Display Top N Only

1. Hold Ctrl (or Cmd on Mac), click and drag the Restaurant Rank pill from the Rows shelf, and drop it onto the Filters card.
2. In the filter dialog box, select a range of ranks (e.g., set the maximum to 10 to show only the Top 10 restaurants).
3. Click OK.



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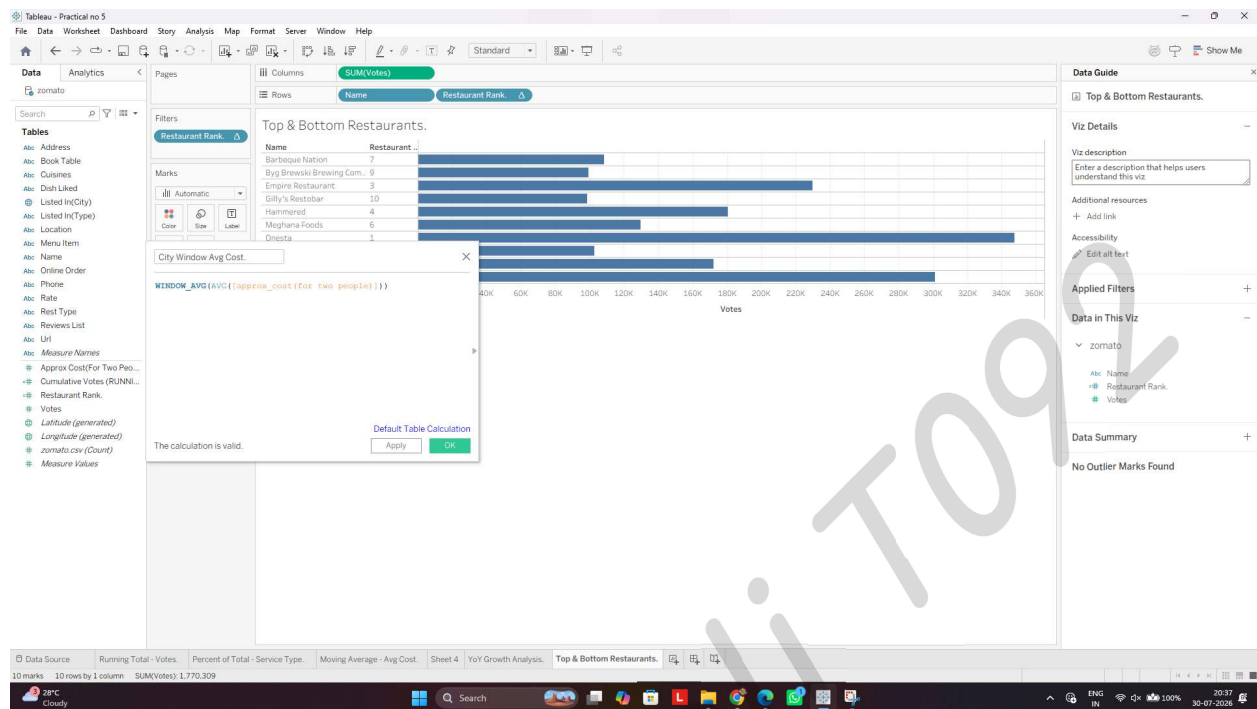


E: Implement window functions (WINDOW_SUM, WINDOW_AVG)

1. Click the small drop-down arrow at the top right of the Data Pane (next to the Search bar) and select Create Calculated Field...
2. Name the field: City Window Avg Cost.
3. Input the following window function formula:
`WINDOW_AVG(AVG([approx_cost(for two people)]))`

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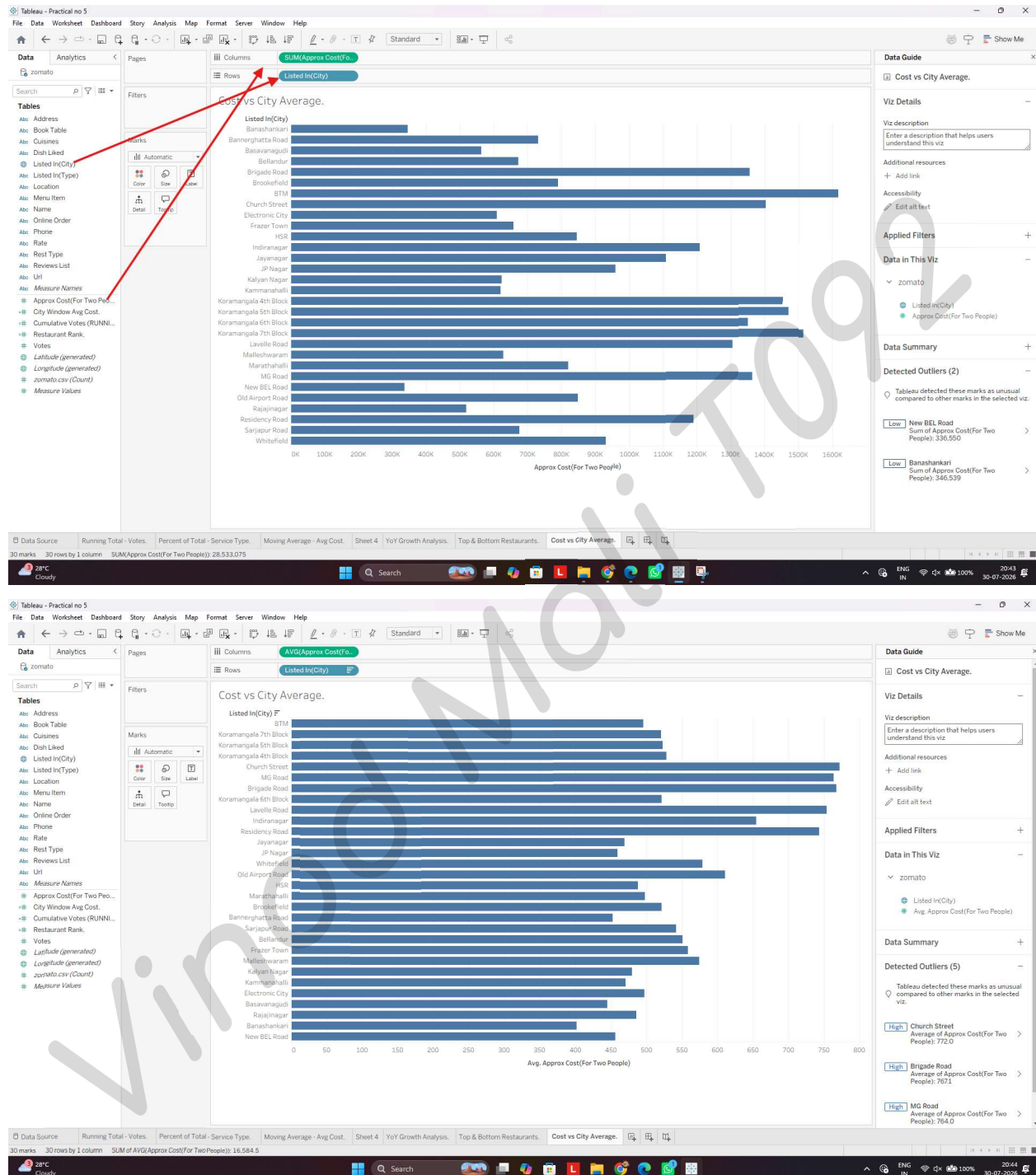
4. Click OK.

Task 2: Build the Window Comparison Visualization

1. Open a new worksheet and rename it: Cost vs City Average.
2. Drag listed_in(city) to the Rows shelf.
3. Drag approx_cost(for two people) to the Columns shelf. Change its aggregation to Average (Right-click pill $\rightarrow$ Measure (Sum) $\rightarrow$ Average).

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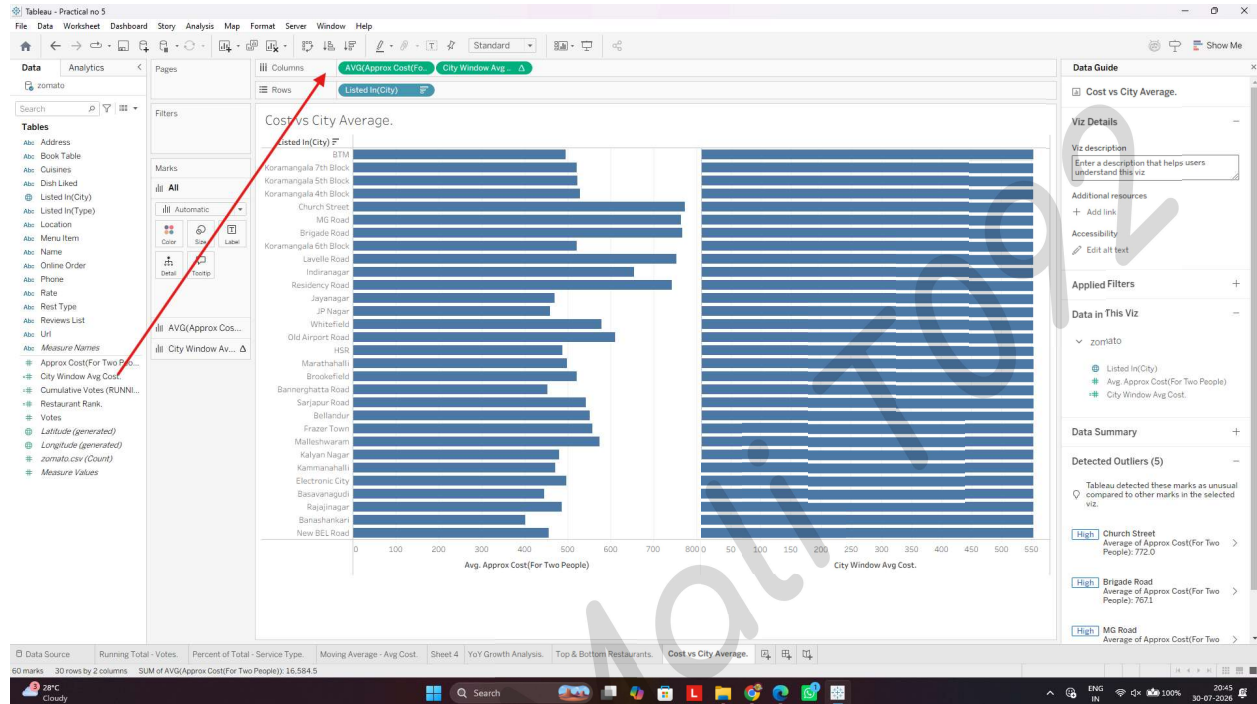
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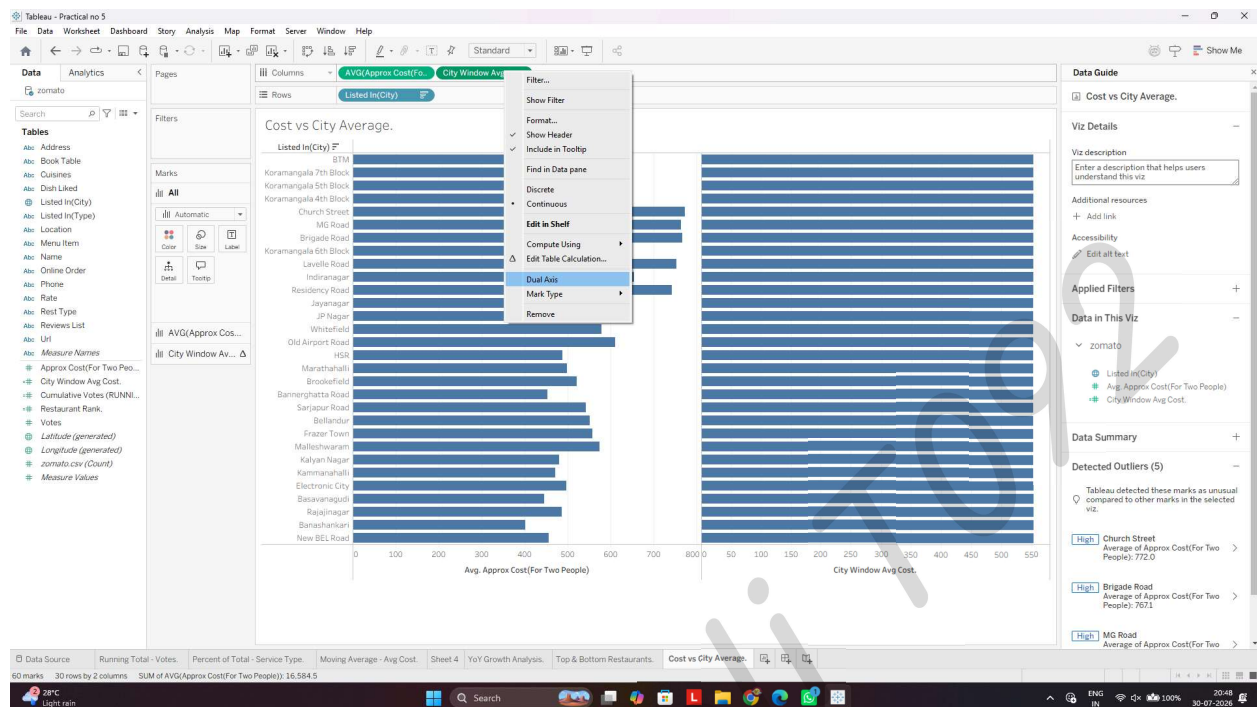
4. Drag your newly created City Window Avg Cost calculated field onto the Columns shelf (placing it right beside AVG(approx_cost...)).



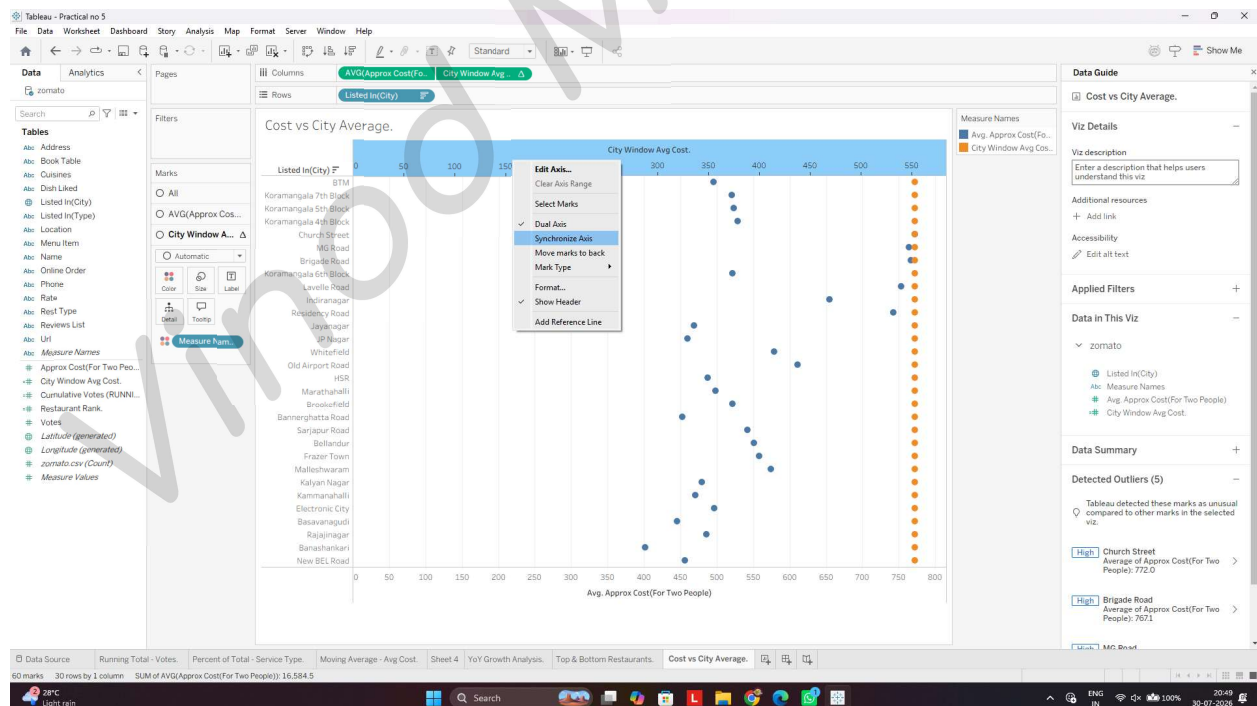
Task 3: Combine Measures into a Dual-Axis Reference View

1. Right-click the City Window Avg Cost pill on the Columns shelf and select Dual Axis.

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2. Right-click the top horizontal axis in the visualization view and select Synchronize Axis.

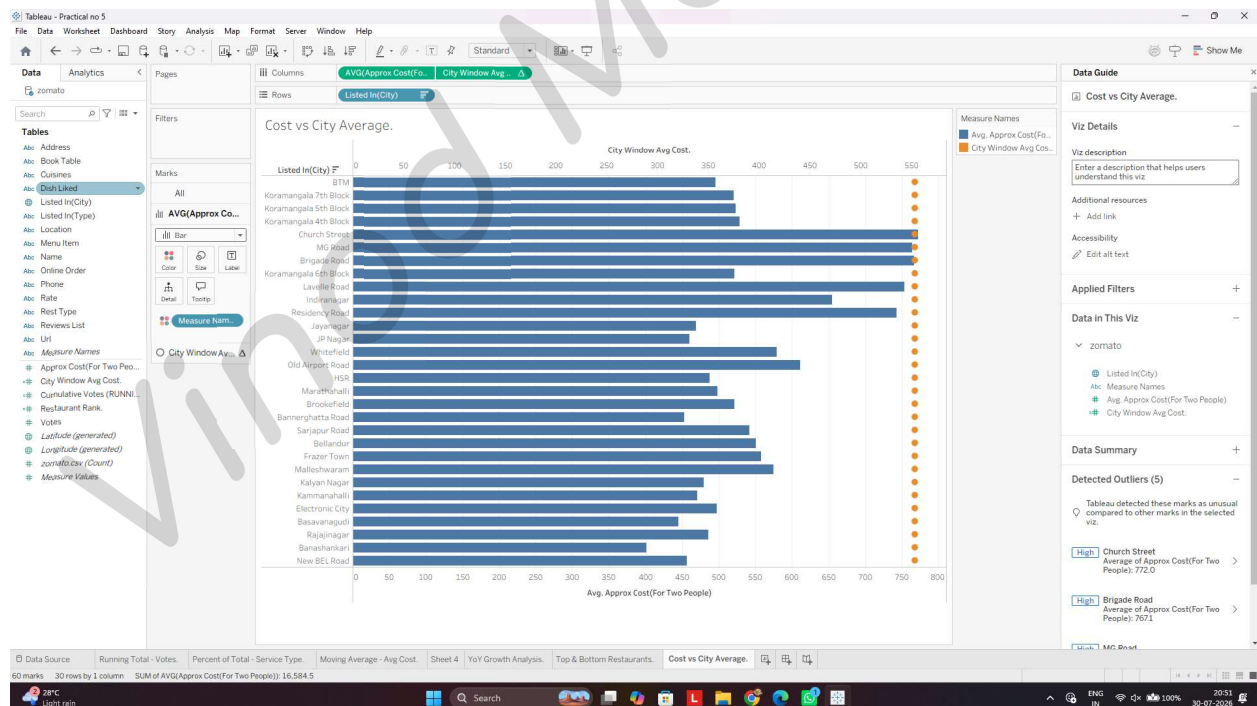
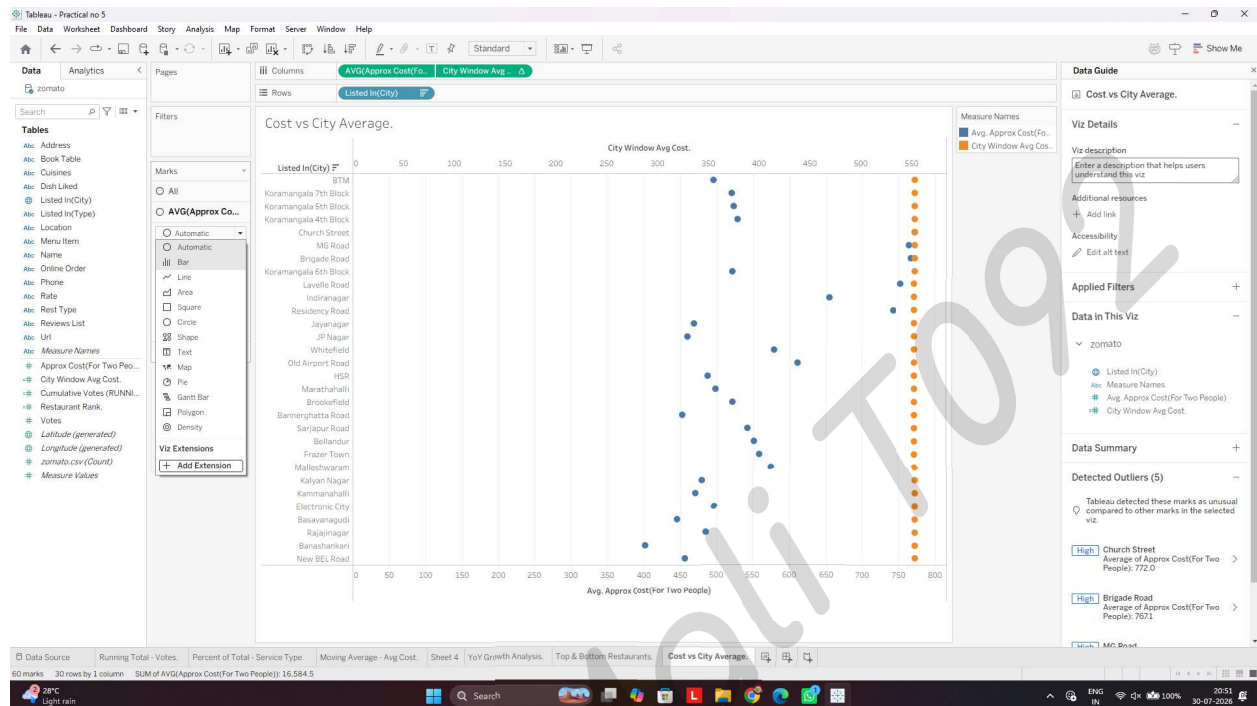


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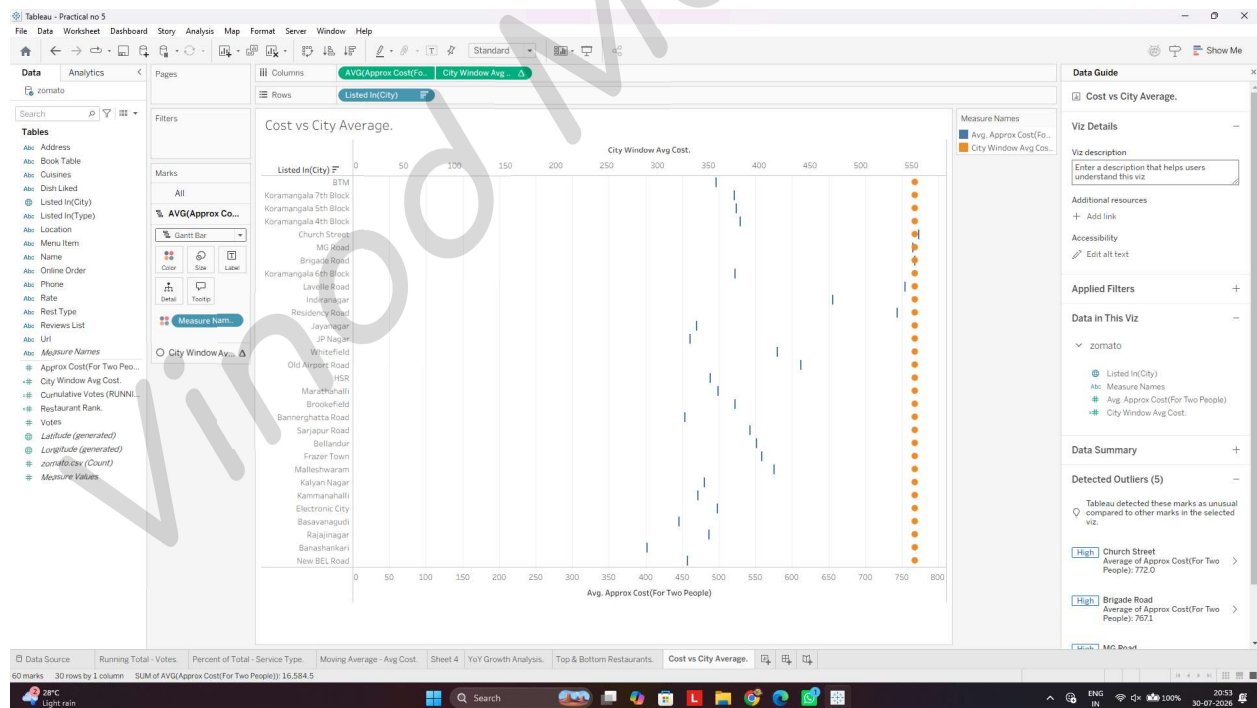
3. In the Marks card:

o Change the mark type for AVG(approx_cost...) to Bar.



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- o Change the mark type for City Window Avg Cost to Line (or Gantt Bar).



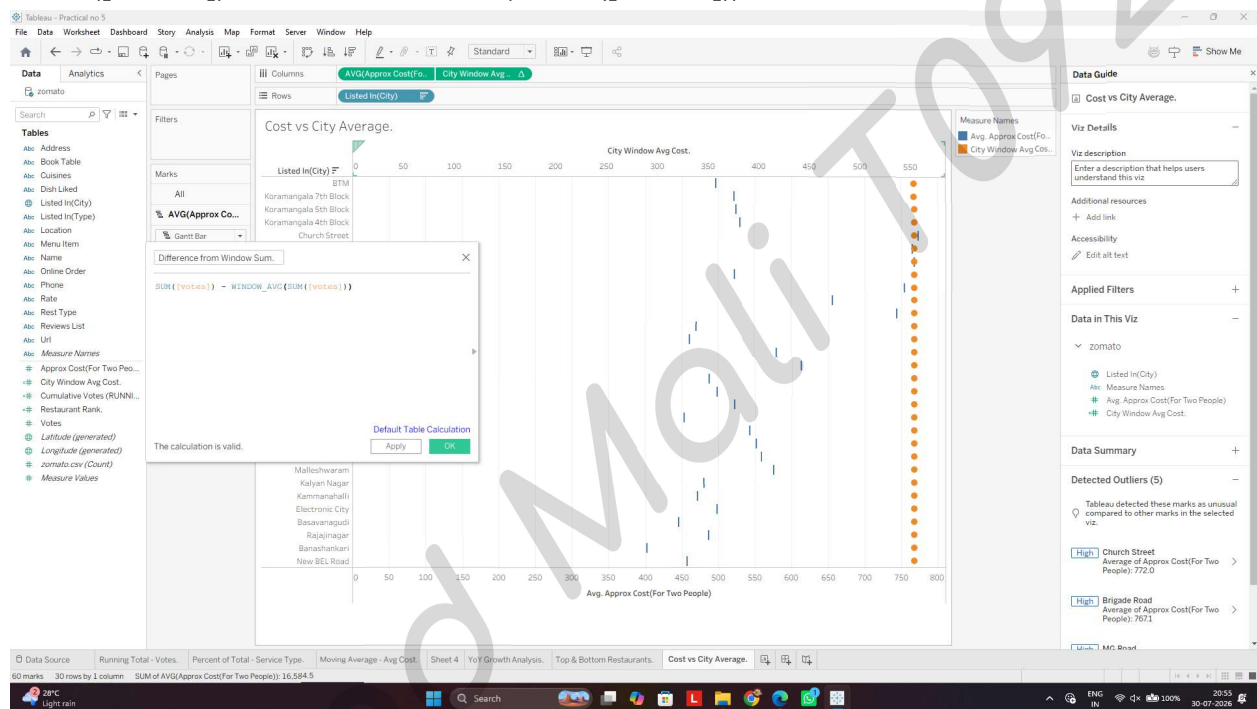
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Task 4: Create a Window Difference Metric (WINDOW_SUM)

1. Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field...
2. Name the field: Difference from Window Sum.
3. Input the following expression to compare each city's vote count against the total sum of all votes in the visible window:

$$\text{SUM}([\text{votes}]) - \text{WINDOW_AVG}(\text{SUM}([\text{votes}]))$$

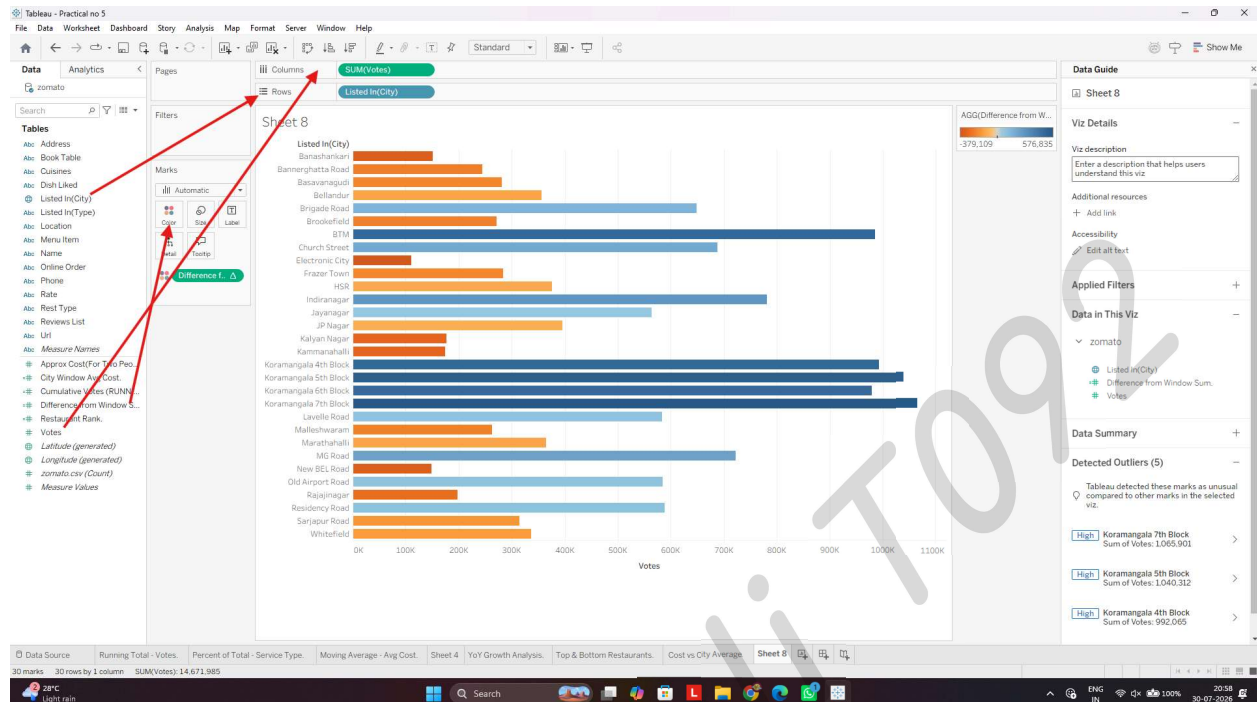


4. Click OK.

5. Apply it to the View:
 - o Open new Sheet
 - o Drag listed_in(city) to the Rows shelf and votes to the Columns shelf.
 - o Drag your new Difference from Window Sum calculated field onto the Color mark on the Marks card.

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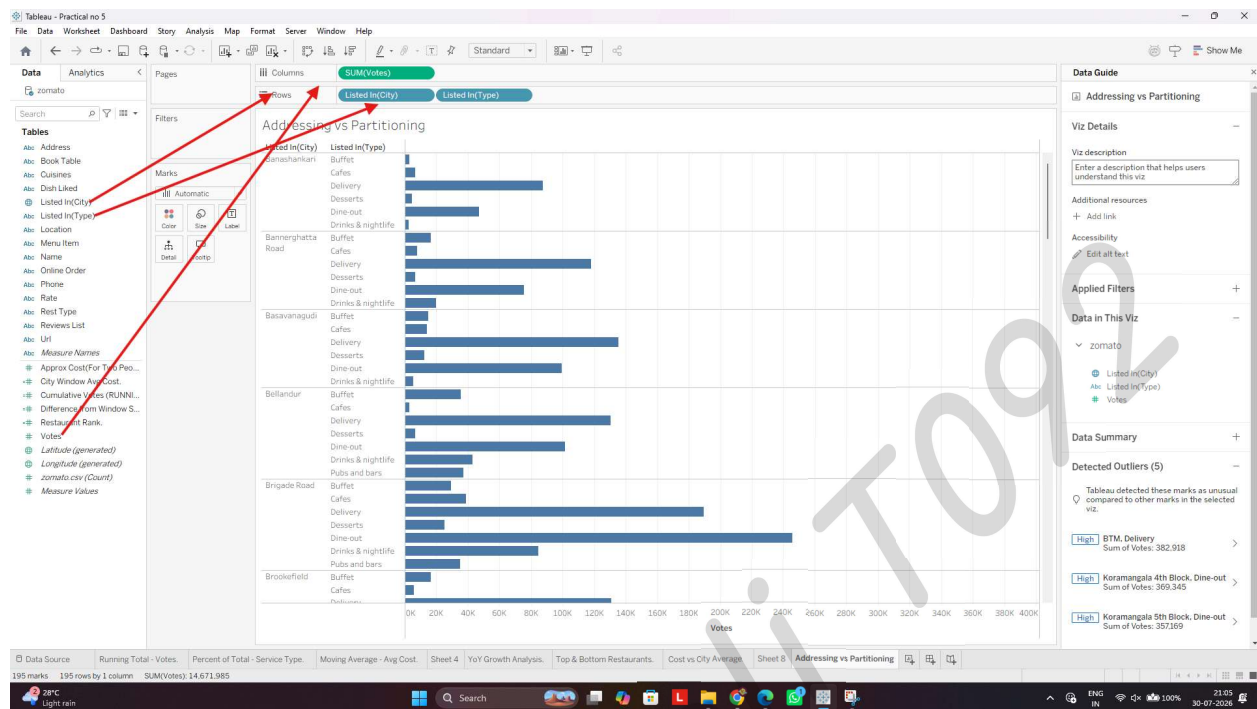


F: Customize addressing and partitioning
Task 1: Build a Nested View for City and Service Type

1. Open a new worksheet and rename it: Addressing vs Partitioning.
2. Drag listed_in(city) to the Rows shelf.
3. Drag listed_in(type) to the Rows shelf, placing it right next to listed_in(city) (creating a two-level nested hierarchy).
4. Drag votes to the Columns shelf as SUM(votes).

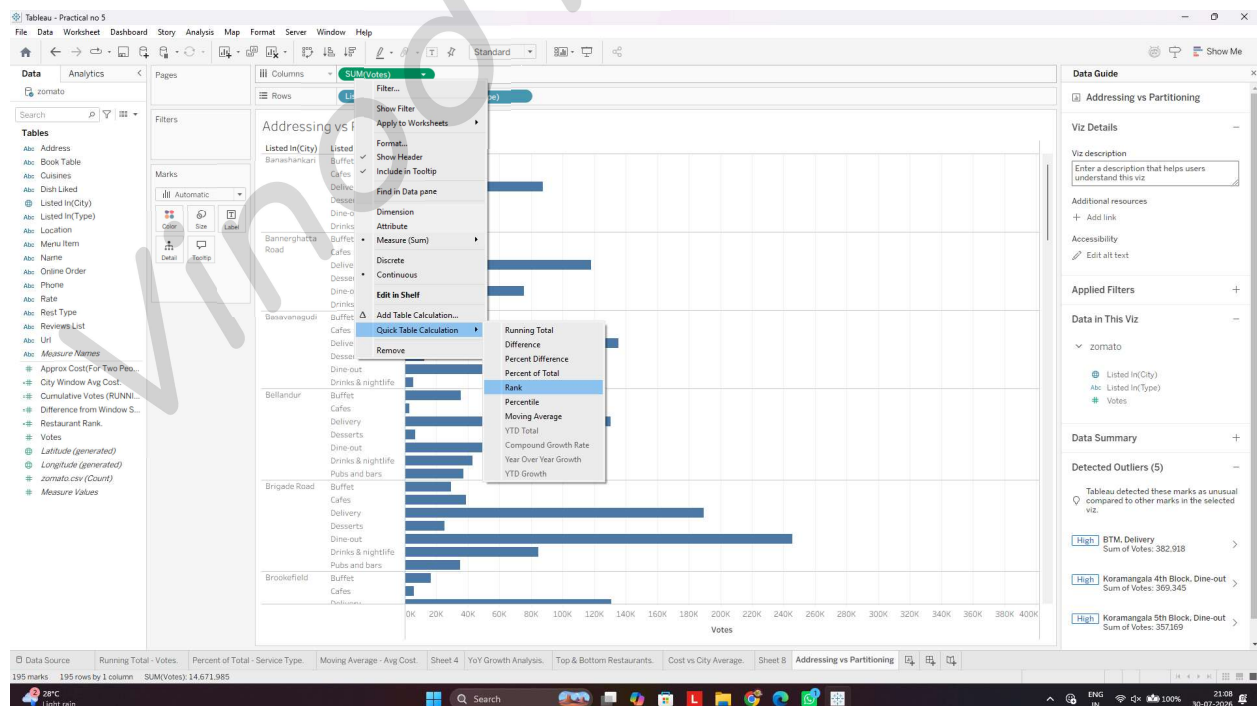
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Task 2: Apply Partitioning via "Compute Using" Options

1. Right-click the SUM(votes) pill on the Columns shelf, hover over Quick Table Calculation, and select Rank.

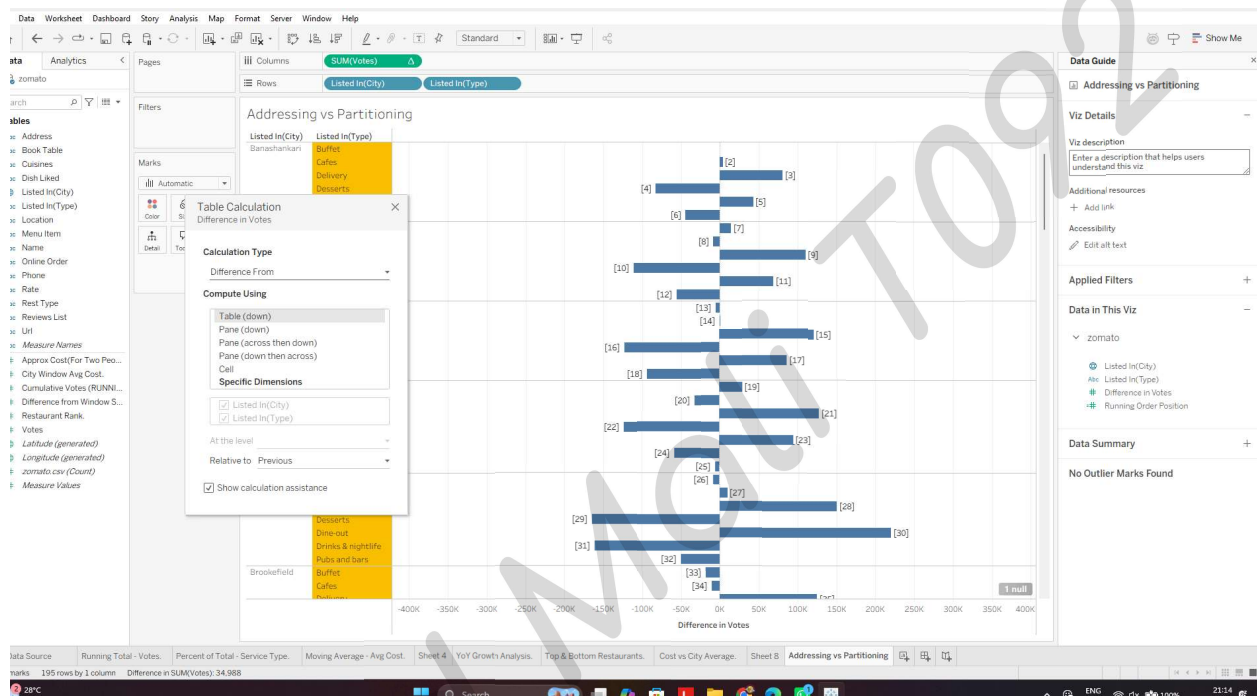


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Subject :- Data Visualization with Power BI and Tableau
Practical No 5

2. Right-click the SUM(votes) pill again and hover over Compute Using:

o Select Table (down): Tableau addresses the entire table from top to bottom. It ranks every single row relative to the whole sheet (Partition = Entire Table).



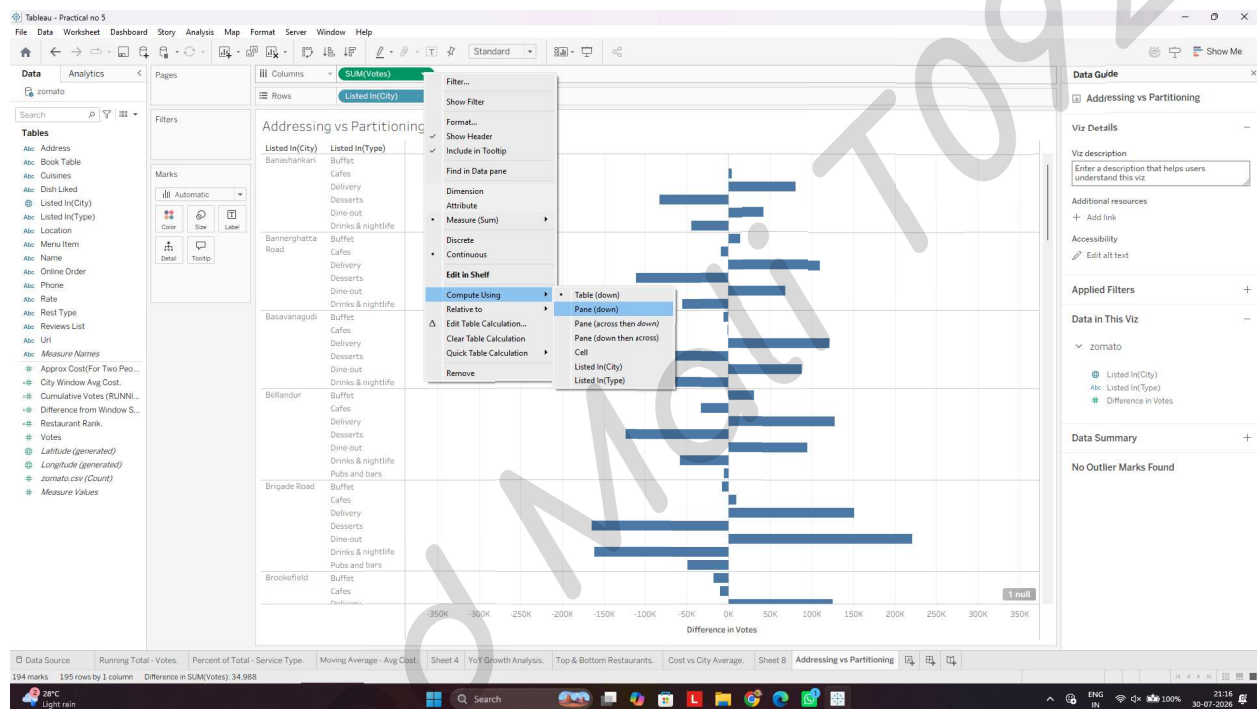
o Select Pane (down): Tableau partitions the calculation by listed_in(city). The ranking now restarts at 1 for the service types inside each individual city!

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Task 3: Manually Define Addressing with Specific Dimensions

For precise control over complex visual structures, use the Specific Dimensions settings:

1. Right-click the SUM(votes) pill (with the Δ icon) on the Columns shelf and select Edit Table Calculation...



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